# Option-Only Portfolio Optimization with Greek-Induced Covariance and Conditional Risk Premia

Dhruv Kohli


### Abstract

How should an allocator choose among listed calls and puts when the instruments are funded, expiring, state-contingent cashflows rather than ordinary asset return columns? This paper builds the option-only analogue of Markowitz for that problem. Each option is represented by premium-funded NAV exposure, payoff and settlement convention, conditional option-risk-premium forecast, Greek exposure, implementation screen, and residual risk. The covariance matrix is induced by the Greek map,

$$r_O = \mu_{O,t} + B_t f_{t+1} + \varepsilon_{t+1}, \qquad \Sigma_{O,t} = B_t \Omega_t B_t^\top + \Sigma_{\varepsilon,t},$$

and the optimizer applies the familiar efficient-frontier algebra to premium weights scaled by portfolio NAV.

In this sample and under these conventions, the empirical evidence is intentionally mixed. The framework is coherent, point-in-time, and auditable. Gross option-only portfolios can look attractive, but gross Sharpe is a research diagnostic, not a trading result. VIX options can improve the design as volatility instruments only when VX-forward modeling and exact VRO/SOQ settlement support the relevant rows. Post-cost accounting, liquidity screens, beta and Greek budgets, inference, and short-option risk controls make the deployable interpretation much more conditional. The portfolio-construction contribution is therefore not a live-trading claim. It is a disciplined way to decide whether convexity, volatility exposure, or short-premium risk deserves capital after premium accounting, costs, settlement, beta audits, Greek budgets, and out-of-sample validation.


### Key Findings

- Option-only Markowitz is feasible when options are treated as premium-funded cashflows with Greek-induced covariance, not as generic return columns.
- Gross option portfolios can appear attractive, but post-cost research accounting, settlement gates, and short-option risk controls sharply limit what can be claimed.
- VIX options are useful as conditional volatility instruments only when VX-forward modeling and exact VRO/SOQ settlement support the evidence.

**Impact Statement**

Option research often begins with a named trade: covered calls, crash puts, variance spreads, or volatility hedges. This paper begins one layer earlier. It gives a reproducible template for turning arbitrary listed calls and puts into NAV-normalized cashflows, Greek exposures, conditional premia, covariance estimates, and constraints before asking whether they deserve capital. The conclusion is useful precisely because it is restrained: option-only Markowitz is mathematically clean, but empirical claims survive only if premium accounting, settlement, costs, inference, and equity-beta audits support them.

## 1. Introduction

An investor restricted to listed options has a simple question with a difficult answer: which calls and puts actually deserve premium capital? The usual asset-allocation template starts from return columns. Options do not fit that template cleanly. They are funded, expiring, state-contingent cashflows whose economics depend on premium paid or received, strike, expiry, exercise style, settlement, volatility, convexity, and the state of the underlying.

This paper builds a premium-weighted option-only Markowitz framework for that choice. The universe can contain arbitrary listed calls and puts across equities, strikes, expiries, and moneyness buckets. Positions can be long or short. Cash is not optimized as a separate risky asset; it is the NAV numeraire and collateral account used to scale option premium exposures. The premium is still paid or received. A long option that expires worthless loses its premium weight, and a short option that finishes in the money pays the payoff through the same NAV accounting. The allocation problem is therefore not "which stock return column has the best mean," but whether a funded option cashflow deserves capital after settlement, costs, beta exposure, Greek budgets, and validation gates.

### *1.1. Research Question and Contributions*

The research question is: how should an allocator choose among listed calls and puts when every instrument is a funded, expiring, state-contingent cashflow rather than a standard asset return column? The proposed answer is a premium-weighted option-only Markowitz framework that maps listed options into NAV-normalized cashflows, payoff and settlement conventions, Greeks, residual risk, conditional option-risk-premium forecasts, implementation screens, and portfolio constraints. In this sample and under these conventions, the validated empirical claim is that the framework is coherent, point-in-time, auditable, and useful for diagnosing whether option convexity, VIX exposure, or short-premium risk deserves capital. The non-claim is equally important: the article does not claim live alpha, production tradability, broker-executed performance, live margin parity, or deployable option trading performance.

The organizing thesis is that option-only portfolio construction should be written as conditional option portfolio optimization. A listed option is not a stock with a different ticker. It is a state-contingent claim with delta, gamma, vega, theta, exercise, settlement, liquidity, and marking conventions. Two options can share the same underlying and still contribute very different risks. A put and a call can leave the book with little net delta and large volatility or convexity exposure. The covariance matrix must therefore come from the instrument map, not only from historical option-price correlations.

The theoretical contribution is the option-only analogue of Markowitz. Classical portfolio theory starts with expected returns and an asset covariance matrix (Markowitz, 1952). This paper starts with option premium weights, a local Greek return map, conditional option-risk-premium forecasts, and a residual covariance term. Once those objects are built, the mean–variance algebra is familiar: the unconstrained maximum-Sharpe direction is proportional to $\widehat{\Sigma}_{O,t}^{-1}\widehat{\mu}_t$. The difference is the construction of $\widehat{\Sigma}_{O,t}$ and $\widehat{\mu}_t$, not the final quadratic geometry.

What is new is the object rather than a single trade. The paper is not a defense of long calls, short puts, covered calls, protective puts, or a named volatility spread. It asks how to allocate across an arbitrary option universe once each contract has a mark, a payoff convention, a Greek exposure vector, a conditional expected return, and an implementation ledger. The expected-return side is also explicit: covariance optimization alone cannot create a high Sharpe ratio. A useful option allocator needs forecastable premia such as carry, variance risk premia, volatility risk premia, skew compensation, tail-insurance pricing, relative value, and regime-dependent convexity.

### *1.2. Related Work and Positioning*

The starting point is mean–variance allocation. Markowitz (1952) showed how expected returns and a covariance matrix define an efficient frontier, while Sharpe (1966) made reward-to-variability central to performance evaluation. Listed options require a different first step because the columns are not ordinary asset returns. Black–Scholes–Merton option valuation (Black and Scholes, 1973; Merton, 1973) and the Black–76 forward option model (Black, 1976) provide local pricing and Greek maps, not a complete physical-measure portfolio allocation rule. The present framework uses those maps to translate options into delta, gamma, vega, theta, VIX-forward exposure, and residual risk before applying Markowitz geometry.

Expected option returns have their own economic sources. Empirical option-return research links compensation to carry, volatility risk premia, skew, crash insurance, and tail-risk states (Coval and Shumway, 2001; Bakshi and Kapadia, 2003; Carr and Wu, 2009; Bollerslev et al., 2009; Broadie et al., 2009). This article uses that spine to motivate conditional option-risk-premium forecasts rather than treating the optimizer as a source of alpha. Robust covariance estimation also matters because inverse-covariance portfolio rules amplify noisy inputs; shrinkage follows Ledoit and Wolf (2004). Evaluation uses both reward-to-volatility and downside-aware measures, including Sortino (Sortino and Price, 1994), Omega (Keating and Shadwick, 2002), Calmar and drawdown (Magdon-Ismail and Atiya, 2004), and active risk (Grinold and Kahn, 2000).

Implementation frictions and validation discipline determine the strength of any empirical claim. Option overlays are sensitive to spread, settlement, borrow, assignment, margin, and capacity assumptions. Backtest overfitting and multiple testing can turn optimizer variants into false discoveries, so the analysis uses bootstrap inference and reality-check ideas consistent with Bailey et al. (2017) and López de Prado (2018). The distinction from strategy-specific option papers is therefore deliberate. The object is not a covered-call, put-write, crash-put, or volatility-spread test. The object is an instrument-level cashflow accounting and allocation framework for arbitrary listed options before asking whether any option sleeve deserves capital.

The empirical result is intentionally mixed. The gross point-in-time backtest tests whether the option-only Markowitz object behaves coherently on the local OPRA-derived panel. The post-cost research layer subtracts conservative spreads, fees, slippage, borrow, capacity, simulated margin funding, and assignment-risk penalties. VIX options are included as volatility derivatives with VX-forward Black–76 Greeks, but VIX option evidence is headline-grade only when exact VRO/SOQ settlement supports the relevant rows. If the coverage table reports proxy rows, the same evidence is reported only as diagnostic hedge-sleeve evidence rather than a listed-settlement claim.

The article proceeds in the same order as the research workflow. Section 2 defines the data and empirical design. Section 3 builds the option cashflow and Greek-risk framework. Section 4 describes the conditional portfolio construction model. Section 5 sets out the metrics, point-in-time validation, inference, and claim gates. Section 6 reports the empirical results and negative evidence. Section 7 concludes with the limits of the current research implementation. Appendix A contains supplemental figures, option/VIX diagnostics, robustness tables, instrument conventions, proof sketches, and reproducibility notes.

## 2. Data

The empirical design is a monthly, point-in-time allocation study over listed equity options and VIX options. The investable equity-option universe is restricted to AAPL, AMZN, GOOGL, JPM, META, MSFT, NVDA, and TSLA, using OPRA/Databento-derived option panels after basic mark, expiry, Greek, and liquidity filters. The VIX-option universe is built from raw VIX chain shards and kept separate because VIX options are volatility derivatives, not options on an equity spot price. Table 1 records the sample size, observation count, train/test split, settlement status, and post-cost layer used in the baseline run.

| Item | Value |
|---|---|
| Raw OPRA-derived equity option rows | 160,315 |
| Raw VIX option rows after filters | 103,650 |
| Monthly snapshot dates | 129 |
| Training dates | 69 |
| Test dates | 60 |
| Primary equity underlyings | AAPL, AMZN, GOOGL, JPM, META, MSFT, NVDA, TSLA |
| VIX option treatment | VX-forward Black-76 Greeks; VIX/VVIX state only |
| VIX settlement source | vro_soq_exact:536 |
| VIX headline status | headline-grade exact VRO/SOQ |
| Post-cost layer | pre-production research simulation: mid, half-spread, and full-spread executable-cost scenarios |
| Broker execution status | not broker-executed live evidence |
| Rolling option bucket assets | 54 |
| Train/test split | through 2020-12-31 / after 2020-12-31 |

**Table 1:** Empirical design summary. The table identifies the option universe, monthly snapshots, train/test split, VIX settlement status, and post-cost convention used in this sample and under these conventions.

The train/test split is frozen at December 31, 2020. Training observations through that date estimate expected option returns, covariance objects, residual risk, factor exposures, and implementation screens. Test observations after that date are reserved for out-of-sample portfolio evaluation. Rebalancing is monthly. At each decision date, the research implementation sees only the option panel, state variables, underlying prices, VX curve, forecast inputs, and cost inputs observable at or before that date. The realized option return is then measured at the next monthly realization date or at listed expiry, according to the holding ledger.

Equity-option rolling buckets convert expiring contracts into stable empirical objects without pretending that options are perpetual assets. For each underlying, right, moneyness, tenor, and liquidity bucket, the decision date selects a representative listed contract using contemporaneous marks and filters. The selected option can be a call or a put, and the optimizer can hold it long or short subject to gross NAV, concentration, Greek, beta, short-premium, liquidity, stress, and assignment-risk constraints. The return includes the premium paid or received: a long option that expires worthless loses its premium weight, while a short option that finishes in the money pays the realized payoff through NAV accounting.

The VIX-option construction follows a different convention. Raw VIX chain shards are stacked, OSI symbols are parsed, duplicate (trade date, symbol) rows are coalesced, and each option is aligned to the relevant VX futures forward. Black–76 Greeks are computed with respect to the VX forward rather than spot VIX. VIX, VVIX, VIX9D, and VIX3M enter only as state variables. Exact VRO/SOQ settlement is a claim gate: a VIX option becomes headline-grade only when the expiry row uses `vro_soq_exact`. Proxy rows, if any, remain appendix diagnostics and cannot support listed-settlement claims.

| Settlement source | Rows | Headline eligible |
|---|---|---|
| vro_soq_exact | 536 | yes |

**Table 2:** VIX settlement coverage. VIX option evidence is headline-grade only when every VIX expiry row used in headline tables has `vro_soq_exact` settlement.

Marks, payoffs, settlement, and costs are treated as first-order design choices. Option weights are premium exposures per dollar of NAV. The mark source, payoff date, settlement source, Greek model, state snapshot, and train-end date are recorded before performance is reported. Post-cost research returns use three executable-cost scenarios, explicit fees, spread haircuts, no-fill rules, capacity limits, simulated margin/stress capital, borrow proxies, and assignment-risk flags. These conventions are adequate for a research paper on allocation mechanics because they test whether the option-only Markowitz object can be constructed, audited, stress-screened, and evaluated out of sample. They are not adequate for production trading claims because they do not contain broker-routed orders, live quote depth, fill probabilities calibrated from orders, live margin previews, or broker reconciliation.

| Object | Accounting rule | Main failure if omitted |
|---|---|---|
| Equity option | Premium weight times mark or payoff return | Treats the option as free convexity |
| Short option | Premium received plus future liability | Hides tail and assignment exposure |
| VIX option | Black–76 exposure to matched VX forward | Treats spot VIX as tradable |
| Cash/collateral | NAV numeraire, not optimized asset | Converts the model into a stock-cash allocation |
| Post-cost layer | Spreads, fees, slippage, borrow, capacity, margin, assignment flags | Promotes gross research returns to false trading evidence |

**Table 3:** Option accounting contract. Each object is represented by the cashflow convention that must be specified before the optimizer can compare it with another option.

## 3. Option Cashflow and Risk Framework

The central modeling step is to express every option as a NAV-normalized cashflow. Let $W_t$ be beginning portfolio NAV, $q_{i,t}$ the signed premium weight assigned to option $i$, and $C_{i,t} > 0$ the decision-date mark. The contract count is

$$N_{i,t} = \frac{q_{i,t} W_t}{C_{i,t}}.$$

Ignoring transaction costs and slippage for the gross theory object, but not ignoring the option purchase price, the next-period NAV contribution is

$$\frac{\Delta W_{i,t+1}}{W_t} = q_{i,t} \frac{C_{i,t+1} - C_{i,t}}{C_{i,t}}.$$

These are premium weights. A long option whose mark falls to zero loses exactly the paid premium weight. A short option is scaled by the premium it sold and then pays the future mark or payoff liability.

For a call or put on equity underlying $u$, the local mark change is approximated by the same Ito–Taylor accounting that underlies Black–Scholes–Merton option valuation (Black and Scholes, 1973; Merton, 1973; Shreve, 2004):

$$\Delta C_i \approx \Delta_i \Delta S_u + \frac{1}{2} \Gamma_i (\Delta S_u)^2 + \nu_i \Delta \sigma_i + \Theta_i \Delta t + \eta_i.$$

The paper does not require Black–Scholes to be a globally correct pricing model for American single-name options. It uses the weaker local statement: an option mark can be expanded in spot, volatility, and time, with residual marking error left in $\eta_i$.

Dividing by the option mark gives the return exposure map

$$r_i = \underbrace{\frac{\Delta_i S_u}{C_i}}_{b^{\Delta}_{i,u}} R_u + \underbrace{\frac{1}{2} \frac{\Gamma_i S_u^2}{C_i}}_{b^{\Gamma}_{i,u}} R_u^2 + \underbrace{\frac{\nu_i}{C_i}}_{b^{\nu}_{i,u}} \Delta \sigma_i + \underbrace{\frac{\Theta_i \Delta t}{C_i}}_{\theta^*_i} + \epsilon_i.$$

The coefficients are dimensionless option-return exposures. Deep out-of-the-money options can have large return exposure because the premium denominator is small. That is not a numerical nuisance. It is the reason option-only portfolios need explicit gross, per-contract, Greek, beta, and stress budgets.

Stacking contracts gives the factor form

$$r_O = \mu_{O,t} + B_t f_{t+1} + \varepsilon_{t+1}, \qquad \Sigma_{O,t} = B_t \Omega_t B_t^{\top} + \Sigma_{\varepsilon,t}.$$

Here $f$ contains underlying returns, centered squared returns, and implied-volatility shocks; $B_t$ contains delta-return, gamma-return, and vega-return loadings; and $\Sigma_{\varepsilon,t}$ is the residual option-return covariance. The residual term is part of the model. Omitting it would assume that Greeks span all marking risk.

VIX options require a separate convention. They are not options on an equity spot. The traded forward-risk anchor is the matched VX futures curve, and Black's futures-option logic is the natural local model (Black, 1976). For VIX option $j$, the local approximation is

$$\Delta C_j^{VIX} \approx \Delta_j^{VX} \Delta F_{T_j}^{VX} + \frac{1}{2}\Gamma_j^{VX}(\Delta F_{T_j}^{VX})^2 + \nu_j^{VIX} \Delta\sigma_j^{VIX} + \Theta_j^{VIX} \Delta t.$$

VIX, VVIX, and term-structure indices are useful state variables; VX futures and VIX options are the traded volatility instruments. Exact VRO/SOQ settlement is required before VIX option expiry P&L can be promoted from settlement proxy to headline listed-settlement evidence.

## 4. Portfolio Construction Model

The optimizer is chosen only after the cashflow and risk map are built. Covariance is not enough. A high Sharpe ratio cannot come from $\Sigma_{O,t}$ alone; it must come from a conditional expected-return vector. The paper writes the option mean as

$$\widehat{\mu}_{i,t} = b_{i,t}^S \widehat{\lambda}_t^S + b_{i,t}^\Gamma \widehat{\lambda}_t^{var} + b_{i,t}^\nu \widehat{\lambda}_t^{vol} + b_{i,t}^{skew} \widehat{\lambda}_t^{skew} + b_{i,t}^{tail} \widehat{\lambda}_t^{tail} + \theta_{i,t}^*.$$

Delta loads on equity compensation. Gamma and vega load on variance and volatility risk premia. Put skew and VIX call wings load on crash-insurance and tail premia. Theta is carry under unchanged state variables. The option-return literature documents why these components can carry expected returns rather than only hedge noise (Coval and Shumway, 2001; Bakshi and Kapadia, 2003; Carr and Wu, 2009; Bollerslev et al., 2009; Broadie et al., 2009). In implementation the combined signal is estimated only from training-window or decision-date information and is shrunk toward zero.

Given $\widehat{\mu}_t$ and $\widehat{\Sigma}_t$, the gross option-only Sharpe problem is

$$\max_{q \in \mathcal{K}_t} \frac{q^\top \widehat{\mu}_t}{\sqrt{q^\top \widehat{\Sigma}_t q}}, \qquad \widehat{\Sigma}_t = B_t \Omega_t B_t^\top + \Sigma_{\varepsilon,t}.$$

If the feasible set is unconstrained and $\widehat{\Sigma}_t$ is positive definite, the maximum-Sharpe direction is proportional to $\widehat{\Sigma}_t^{-1}\widehat{\mu}_t$. This is exactly the Markowitz tangency formula. The option work is in building the conditional option return object that the formula consumes.

The practical feasible set is smaller. The research implementation imposes gross NAV, short premium, per-contract, underlying concentration, delta, SPY beta, eight-underlying beta, gamma, equity vega, VIX vega, and scenario stress-loss budgets where the corresponding strategy requires them. The optimizer can therefore choose calls, puts, long positions, and short positions, but it cannot hide large exposure behind a small option premium.

| Strategy | Predicted monthly vol | Train realized vol | Test realized vol | Test / predicted |
|---|---|---|---|---|
| Equity-option Greek Markowitz | 0.334 | 0.267 | 0.307 | 0.918 |
| Greek Markowitz + VIX | 0.363 | 0.261 | 0.262 | 0.722 |
| Beta/delta-neutral + VIX | 0.354 | 0.272 | 0.239 | 0.677 |
| Equal premium | 1.509 | 0.625 | 0.542 | 0.359 |
| Equal risk | 0.668 | 0.394 | 0.368 | 0.551 |
| VIX hedge sleeve | 4.648 | 4.092 | 0.417 | 0.090 |

**Table 4:** Greek-induced covariance calibration. Predicted monthly volatility is computed from $q^\top \Sigma_O q$ using the training-window Greek risk model; realized volatility is measured on train and test option returns.

The estimator follows the standard risk-model route. It constructs $B_t$, estimates $\Omega_t$ from point-in-time state shocks, estimates $\Sigma_{\varepsilon,t}$ from training residuals, shrinks the covariance toward a diagonal target, and repairs small negative eigenvalues. This follows the same concern that motivates covariance shrinkage in ordinary portfolio problems: noisy matrices become dangerous when the optimizer inverts them (Ledoit and Wolf, 2004).

## 5. Metrics and Validation

The validation design asks several different questions. The realized path asks what happened in the held-out monthly window. The point-in-time audit asks whether decisions used only information available at the rebalance date. The regression asks whether the option book is mostly equity drift or volatility exposure. The attribution asks whether P&L comes from delta, gamma, vega, carry, VIX-forward exposure, skew/tail terms, or residual marking error. Cost stress asks whether the gross result survives realistic research frictions.

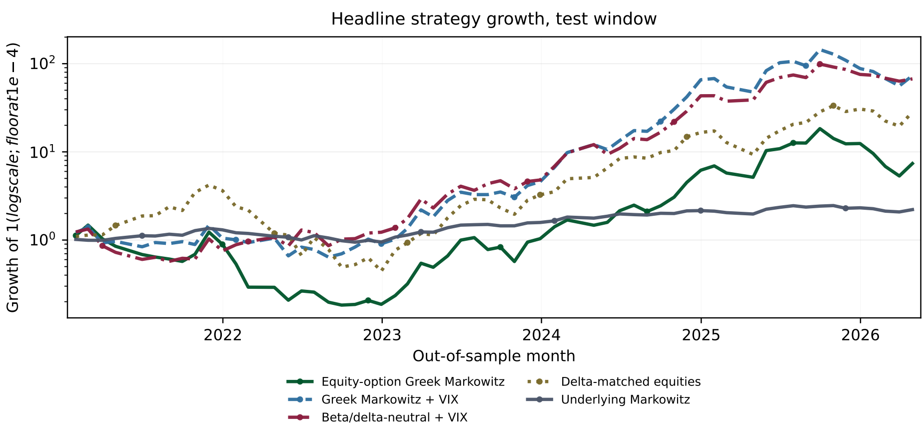


**Figure 1:** Headline out-of-sample growth paths. The figure focuses on the central option allocators and equity benchmarks; the all-strategy growth diagnostic appears in Appendix A. The log-scale floor prevents failed strategies from disappearing visually.

Performance is reported with complementary measures. Sharpe follows the reward-to-variability tradition (Sharpe, 1966). Sortino replaces total volatility with downside deviation (Sortino and Price, 1994). Calmar scales return by maximum drawdown (Magdon-Ismail and

Atiya, 2004). Omega compares gains above a threshold with shortfalls below it (Keating and Shadwick, 2002). Information ratio measures active return against SPY (Grinold and Kahn, 2000). The zero monthly threshold is used for Sortino and Omega because cash is a numeraire in the model, not an alpha sleeve.

The inference layer is intentionally conservative. Sharpe, Sortino, Calmar, Omega, regime statistics, and leave-one-out differences are reported with block-bootstrap intervals where meaningful. Factor alpha uses HAC/Newey–West standard errors. These intervals do not prove economic significance, but they prevent the paper from reading point estimates as certainties. The pre-production extension also reports probabilistic Sharpe ratios, deflated Sharpe ratios, and a block-bootstrap max-Sharpe screen across the tested gross and executable-cost variants.

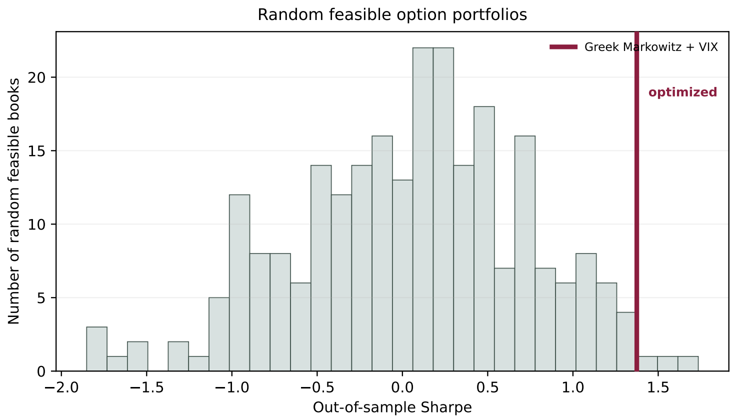


**Figure 2:** Random feasible comparison. The histogram shows out-of-sample Sharpe ratios for random gross-NAV-feasible option books; the vertical marker shows the optimized Greek-Markowitz-plus-VIX sleeve. The exhibit asks whether optimization adds information beyond arbitrary feasible option exposure.

The point-in-time contract is strict. Decision dates precede return dates. State snapshots, Greeks, forecasts, covariances, residuals, betas, borrow proxies, and cost inputs are drawn from the training window or the decision date. Expiry spot, future IV, future VIX/VVIX, future VX curves, and future option returns are not allowed to enter portfolio construction. Cost-scenario ledgers, required-capital denominators, hurdle/no-trade records, liquidity-tier diagnostics, assignment-risk filters, and reality-check inputs obey the same rule. The audit layer reads the machine-readable outputs independently and fails if core lineage, settlement, cost, inference, claim, or manuscript checks fail.

## 6. Results and Discussion

The strongest result is conceptual rather than promotional. In this sample and under these conventions, the option-only Markowitz object is coherent, auditable, and measurable. The empirical investment result is more conditional. Gross option portfolios can look attractive, but gross Sharpe is a research diagnostic rather than a trading result. VIX options improve the design as volatility instruments and hedge candidates only when exact VRO/SOQ settlement coverage supports the headline rows.

The body keeps the central evidence in view: out-of-sample performance, full-spread post-cost survival, factor exposure, the realized path in Figure 1, the random feasible comparison in Figure 2, the VIX settlement gate in Table 2, claim strength, and the claim audit. Detailed liquidity, attribution, regime, forecast-ablation, and multiple-testing diagnostics appear in Appendix A.

### 6.1. Headline evidence

| Return basis | Strategy | Ann. return | Ann. vol | Sharpe | Sortino | Calmar | Omega | Info. ratio |
|---|---|---|---|---|---|---|---|---|
| Gross before costs | Equity-option Greek Markowitz | 0.894 | 1.062 | 0.842 | 1.770 | 1.020 | 1.888 | 0.598 |
| Gross before costs | Greek Markowitz + VIX | 1.246 | 0.907 | 1.374 | 3.385 | 2.034 | 2.784 | 1.200 |
| Gross before costs | Beta/delta-neutral + VIX | 1.172 | 0.829 | 1.414 | 3.410 | 2.057 | 2.948 | 1.147 |
| Gross before costs | Equal premium | -0.056 | 1.878 | -0.030 | -0.055 | -0.056 | 0.977 | -0.120 |
| Gross before costs | Equal risk | 0.104 | 1.275 | 0.082 | 0.146 | 0.107 | 1.064 | -0.058 |
| Gross before costs | VIX hedge sleeve | -9.285 | 1.444 | -6.429 | -3.089 | | 0.040 | -6.132 |
| Gross before costs | Delta-matched equities | 1.095 | 0.931 | 1.176 | 2.317 | 1.225 | 2.367 | 1.041 |
| Gross before costs | Underlying Markowitz | 0.179 | 0.196 | 0.909 | 1.856 | 0.590 | 1.929 | 0.072 |
| Post-cost research | Equity-option Greek Markowitz | -7.900 | 5.476 | -1.443 | -1.347 | -6.357 | 0.098 | -1.545 |
| Post-cost research | Greek Markowitz + VIX | -6.079 | 4.827 | -1.259 | -1.198 | -5.136 | 0.116 | -1.345 |
| Post-cost research | Beta/delta-neutral + VIX | -3.896 | 2.841 | -1.371 | -1.300 | -3.895 | 0.151 | -1.508 |
| Post-cost research | Equal premium | -3.165 | 2.378 | -1.331 | -1.502 | -3.051 | 0.359 | -1.372 |
| Post-cost research | Equal risk | -1.276 | 1.367 | -0.934 | -1.199 | -1.276 | 0.505 | -1.082 |
| Post-cost research | VIX hedge sleeve | -14.364 | 4.209 | -3.413 | -2.445 | -12.391 | 0.016 | -3.357 |
| Post-cost research | Delta-matched equities | 1.095 | 0.931 | 1.176 | 2.317 | 1.225 | 2.367 | 1.041 |
| Post-cost research | Underlying Markowitz | 0.179 | 0.196 | 0.909 | 1.856 | 0.590 | 1.929 | 0.072 |

**Table 5:** Gross and post-cost out-of-sample performance. Gross rows test the option-only Markowitz object; post-cost rows are research simulations that subtract spreads, fees, slippage, borrow, capacity, simulated margin funding, and assignment-risk penalties.

The table separates two questions. The gross rows ask whether the mathematical allocation object works on held-out option returns. The post-cost rows ask whether the same object still looks economically attractive after conservative implementation frictions. The pre-production layer then asks whether the result survives midpoint, half-spread, and full-spread executable-cost scenarios, required-capital denominators, no-fill diagnostics, and margin/stress-capital screens. Gross Sharpe is not treated as a trading result.

| Strategy | Gross Sharpe | Net Sharpe | Net Calmar | Avg spread cost | Capacity used | Survives? |
|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | 0.8421194565895301 | 0.7668067218645858 | 0.9148415644409728 | 0.004454581057950615 | 0.9901002345178692 | yes |
| Greek Markowitz + VIX | 1.3743885779509968 | 1.0258219183556512 | 1.331603783957096 | 0.02189086589088895 | 0.9988451122012506 | yes |
| Beta/delta-neutral + VIX | 1.4135236431740843 | 0.9508576292459209 | 1.2239969998479348 | 0.026468684884072495 | 0.9997049345720702 | yes |
| Equal premium | -0.02955676336526383 | -0.20462197552298175 | -0.38425395579926913 | 0.020340974748224574 | 0.9993803841618195 | no |
| Equal risk | 0.0818308658646486 | -0.11815498556113241 | -0.15161444631136373 | 0.015983091604746466 | 0.9990134907215874 | no |
| VIX hedge sleeve | -6.429184989935395 | -6.607601788466617 | nan | 0.022500000000000003 | 0.9885078059495911 | no |

**Table 6:** Post-cost survival under the full-spread executable-cost scenario. A strategy survives only if net performance remains positive after conservative research frictions, capacity screens, settlement gates, and risk-capital checks.

| Strategy | Ann. alpha | Alpha HAC t | $R^2$ | Residual ann. vol | Beta SPY | Beta VX front | Beta dVIX | Beta dVVIX |
|---|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | 0.405 | 1.141 | 0.618 | 0.634 | 0.005 | -1.368 | 0.046 | 0.003 |
| Greek Markowitz + VIX | 0.675 | 2.060 | 0.513 | 0.616 | 1.908 | -0.767 | 0.056 | -0.001 |
| Beta/delta-neutral + VIX | 0.845 | 2.753 | 0.429 | 0.601 | 0.715 | -0.522 | 0.038 | -0.001 |
| Equal premium | -0.416 | -0.566 | 0.253 | 1.647 | 2.522 | -0.749 | 0.032 | -0.001 |
| Equal risk | -0.183 | -0.334 | 0.253 | 1.115 | 2.041 | -0.379 | 0.020 | -0.003 |
| VIX hedge sleeve | -8.807 | -14.623 | 0.252 | 1.264 | 3.562 | -1.180 | 0.093 | -0.013 |
| Delta-matched equities | 0.000 | 2.229 | 1.000 | 0.000 | -0.000 | -0.000 | -0.000 | 0.000 |
| Underlying Markowitz | 0.000 | 2.700 | 1.000 | 0.000 | -0.000 | -0.000 | -0.000 | 0.000 |

**Table 7:** Headline factor regressions. The table asks whether option returns are explained by equity drift or volatility state controls before any alpha language is considered; the full eight-underlying regression appears in Appendix A.

The regression is an audit, not a causal proof. If the option book is only a levered large-cap equity book, SPY and single-name betas should explain it. If it is mainly a volatility trade, VX, VIX, and VVIX controls should reveal that. A high point estimate without credible inference, low exposure, and post-cost robustness is not promoted to an option-alpha claim.

| Strength | Claim | Boundary |
|---|---|---|
| Strong | Option cashflow accounting, Greek-induced covariance, premium-weighted Markowitz formulation, and validation discipline are coherent. | Mathematical construction and reproducibility checks. |
| Moderate | The point-in-time research implementation produces useful gross and post-cost diagnostics in this sample. | Monthly OPRA/Databento-derived panels, exact-settlement rows, and research cost assumptions. |
| Weak/conditional | Convexity, VIX exposure, and short-premium sleeves help only in selected states after premium, skew, hedge cost, settlement, and risk-budget checks. | State- and settlement-dependent; exact VRO/SOQ rows present. |
| No claim | Live alpha, production tradability, executable capacity, broker-realistic fills, or live margin parity. | No broker-routed orders, live fills, live margin preview, or broker reconciliation. |

**Table 8:** Claim-strength summary. The table distinguishes durable framework claims from sample-dependent diagnostics and from live-trading claims that are not made.

| Claim | Type | Evidence | Status |
|---|---|---|---|
| Option covariance equals $B\Omega B^{\top} + \Sigma_{\varepsilon}$ | Theorem | Displayed proof and standard covariance algebra | Proved |
| Unconstrained option tangency is $\Sigma_O^{-1}\mu_O$ | Theorem | Displayed Lagrange multiplier proof | Proved |
| Greek Markowitz Sharpe exceeds random feasible p95 | Generated empirical | performance table and random feasible Sharpe artifact | Not supported after exact-expiry PIT audit |
| Result is not only long-call equity drift | Generated diagnostic | exposure, regression, regimes, equity benchmarks, leave-one-out | Not supported as an alpha-independence claim |
| Option premium cost is included in P&L and Sharpe | Accounting identity | NAV weights, mark-return construction, exposure premium columns | Implemented by construction |
| Empirical returns use point-in-time option choices | Generated diagnostic | timing diagnostics, trading-data audit, and holding-return detail artifacts | Checked against train/test dates |
| Stock splits do not create option windfalls | Generated diagnostic | raw-close split factors and exact-expiry payoff construction | Checked for detected split/unit jumps |
| Greek risk model fully explains option P&L | Rejected overclaim | approximation diagnostics and P&L attribution artifacts | Not claimed |
| Post-cost research returns include implementation frictions | Generated empirical | Generated mid, half-spread, full-spread, fee, capacity, required-capital, and assignment-risk ledgers | Implemented as conservative research simulation |
| Pre-production results are broker-executed live evidence | Rejected overclaim | No live fills, order routing, broker margin preview, or broker reconciliation | Not claimed |
| VIX option results are headline-grade listed-settlement evidence | Conditional empirical | All VIX headline rows use exact VRO/SOQ source flags | Supported |
| Strategy is production tradable after costs | Production claim | No live fills, live broker margin preview, order routing, or broker reconciliation | Not claimed |
| VIX options are modeled as volatility derivatives, not equity underlyings | Modeling convention | VIX option panel uses VX-forward Black-76 Greeks and VIX/VVIX state variables | Implemented |
| VIX option expiry P&L is exact listed settlement P&L | Generated empirical | All VIX expiry rows use exact Cboe VRO/SOQ settlement | Supported |

**Table 9:** Claim-audit summary. The table separates mathematical claims, empirical diagnostics, settlement-gated VIX claims, post-cost research claims, and production claims that are not claimed.

The claim audit is part of the result. The article can claim a mathematically explicit option-only efficient frontier and a point-in-time empirical implementation. It can report gross, exact-settlement VIX, and pre-production post-cost research diagnostics when the audit layer supports those outputs. It cannot claim production tradability, live fills, live broker margin parity, or executable capacity because those controls are not broker-run in this article.

## 7. Conclusions

The durable contribution is the option cashflow and risk-accounting framework. A listed option enters the allocation problem only after it has been represented as a premium-funded cashflow, marked under a settlement convention, mapped into delta, gamma, vega, theta, VIX-forward exposure, and residual risk, and forecast under the physical measure. Once those objects exist, the Markowitz algebra is familiar: expected option cashflows and a Greek-induced covariance matrix replace ordinary asset means and covariances.

The validation layer supports a coherent point-in-time research implementation, not production tradability. Gross Sharpe, VIX hedge behavior, and option convexity are not enough. They must survive premium accounting, exact settlement controls, post-cost research accounting, beta and Greek attribution, inference, leave-one-out tests, and no-lookahead checks. VIX options are useful in the framework because they are volatility instruments rather than equity substitutes, but their expiry P&L becomes headline-grade only when exact VRO/SOQ settlement supports the rows being reported.

The trader's use case is a disciplined screening rule rather than an automatic trading signal. A sleeve enters the book only when forecasted option premia survive premium cost, spread, settlement, and Greek-budget checks. Convexity is worth paying for only when state variables and hedge diagnostics justify the premium. Short option exposure is rejected or capped when stress loss, assignment risk, liquidity, or margin screens dominate expected carry. VIX options are treated as volatility instruments. When the validation layer cannot separate option premia from equity drift, volatility carry, or optimistic execution, the portfolio should sit in cash collateral, trade smaller, or use simpler diversifiers.

This paper does not claim live-trading performance. The empirical sample is limited to the available OPRA/Databento-derived equity-option panels, VIX option chains, monthly snapshots, and exact VRO/SOQ coverage. Options execution, rates, margin, borrow, assignment, and capacity are modeled through conservative research assumptions rather than broker-level order and fill data. Production deployment would require richer quote histories, executable bid/ask depth, broker margin previews, assignment and exercise modeling, live routing, and reconciliation.

The conclusion is therefore deliberately restrained. Build the option cashflow. Charge the premium and the implementation frictions. Estimate conditional premia before optimizing. Constrain Greeks, beta, VIX exposure, and stress loss. Use regressions, attribution, regimes, leave-one-out tests, and inference to ask whether the result is more than equity drift or volatility carry. Reject any claim that depends on free options, future information, unpriced settlement, optimistic costs, or hidden short-option risk.

## Data availability statement

The reproduction package contains code, generated tables, figures, verification outputs, schemas, and artifact hashes. Raw licensed OPRA/Databento data are not redistributed. Users with the necessary data rights and API keys can rebuild the local feature panels with the documented data-pull commands and environment template. Exact VRO/SOQ files, when provided locally, are versioned by source hash before VIX option rows become headline-grade.

## References

D. H. Bailey, J. M. Borwein, M. López de Prado, and Q. J. Zhu. The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39–69, 2017.

G. Bakshi and N. Kapadia. Delta-hedged gains and the negative market volatility risk premium. *The Review of Financial Studies*, 16(2):527–566, 2003.

F. Black. The pricing of commodity contracts. *Journal of Financial Economics*, 3(1–2): 167–179, 1976.

F. Black and M. Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.

T. Bollerslev, G. Tauchen, and H. Zhou. Expected stock returns and variance risk premia. *The Review of Financial Studies*, 22(11):4463–4492, 2009.

M. Broadie, M. Chernov, and M. Johannes. Understanding index option returns. *The Review of Financial Studies*, 22(11):4493–4529, 2009.

P. Carr and L. Wu. Variance risk premiums. *The Review of Financial Studies*, 22(3): 1311–1341, 2009.

J. D. Coval and T. Shumway. Expected option returns. *The Journal of Finance*, 56(3): 983–1009, 2001.

R. C. Grinold and R. N. Kahn. *Active Portfolio Management*. McGraw-Hill, 2 edition, 2000.

C. Keating and W. F. Shadwick. A universal performance measure. *The Finance Development Centre Working Paper*, 2002.

O. Ledoit and M. Wolf. Honey, i shrunk the sample covariance matrix. *The Journal of Portfolio Management*, 30(4):110–119, 2004.

M. López de Prado. *Advances in Financial Machine Learning*. Wiley, 2018.

M. Magdon-Ismail and A. F. Atiya. Maximum drawdown. *Risk Magazine*, 17(10):99–102, 2004.

H. Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

R. C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

W. F. Sharpe. Mutual fund performance. *The Journal of Business*, 39(1):119–138, 1966.

S. E. Shreve. *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer, 2004.

F. A. Sortino and L. N. Price. Performance measurement in a downside risk framework. *The Journal of Investing*, 3(3):59–64, 1994.

# A. Appendix

**Reproducibility**

Run from the project root after configuring local data paths and credentials where needed:

```
make option-paper
make paper
make verify
make test
```

## *A.1. Additional realized-path and optimizer results*

This supplement begins with realized-path and optimizer diagnostics. The exhibits use the same held-out panel as the body tables and compare allocation behavior rather than re-estimating a separate experiment.

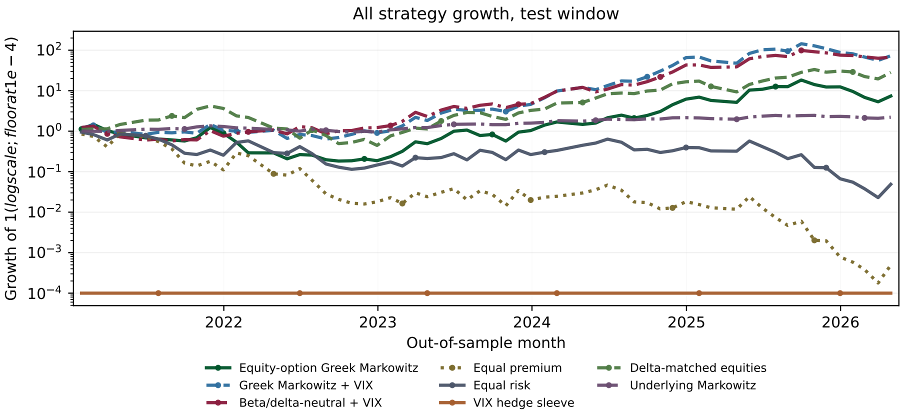


**Figure 3:** All-strategy out-of-sample growth diagnostic. The main text shows the central strategies; this appendix figure keeps every generated strategy visible for auditability.

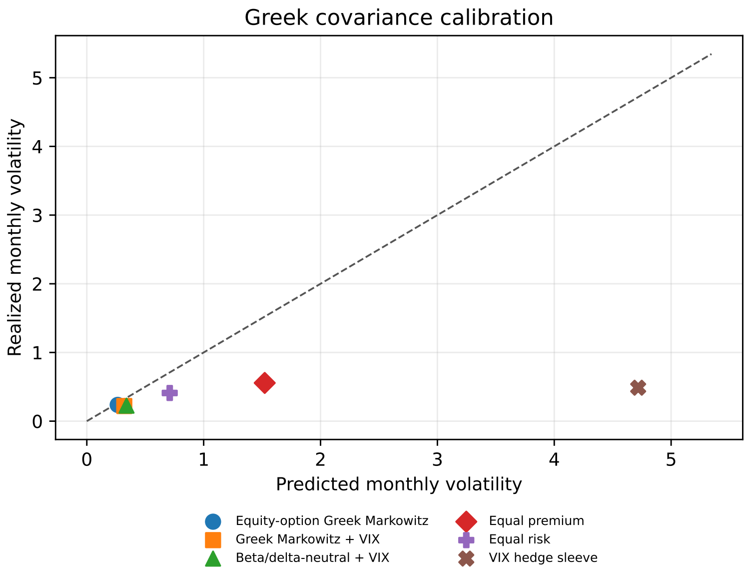


**Figure 4:** Predicted versus realized monthly volatility by option strategy.

| Strategy | Book | Gross NAV | Net NAV | Long premium paid | Short premium sold | Net option premium | Short gross |
|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | Options | 1.000 | 0.500 | 0.750 | 0.250 | 0.500 | 0.250 |
| Greek Markowitz + VIX | Options | 1.000 | 0.500 | 0.750 | 0.250 | 0.500 | 0.250 |
| Beta/delta-neutral + VIX | Options | 1.000 | 0.500 | 0.750 | 0.250 | 0.500 | 0.250 |
| Equal premium | Options | 1.000 | 1.000 | 1.000 | 0.000 | 1.000 | 0.000 |
| Equal risk | Options | 1.000 | 1.000 | 1.000 | 0.000 | 1.000 | 0.000 |
| VIX hedge sleeve | Options | 1.000 | 1.000 | 1.000 | 0.000 | 1.000 | 0.000 |
| Delta-matched equities | Equities | 3.307 | 3.307 | | | | 0.000 |
| Underlying Markowitz | Equities | 1.000 | 0.539 | | | | 0.231 |

**Table 10:** Premium and sign exposures. The table records how much option premium is paid, sold, and netted before the strategy return is realized.

| Strategy | Book | Net delta | Gross delta | Net gamma | Net vega | Equity vega | VIX vega | Beta SPY proxy | Worst stress return | VIX option gross | Call gross | Put gross |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | Options | 2.336 | 4.055 | 5.376 | 0.004 | 0.004 | 0.000 | 3.000 | -0.207 | 0.000 | 0.682 | 0.318 |
| Greek Markowitz + VIX | Options | 3.307 | 3.307 | 5.029 | -0.108 | 0.005 | -0.113 | 2.342 | -0.350 | 0.200 | 0.633 | 0.367 |
| Beta/delta-neutral + VIX | Options | -0.086 | 3.893 | 2.186 | -0.099 | 0.005 | -0.104 | -0.145 | -0.010 | 0.200 | 0.588 | 0.412 |
| Equal premium | Options | 1.208 | 2.202 | 359.583 | 0.211 | 0.041 | 0.170 | 0.897 | 0.258 | 0.093 | 0.537 | 0.463 |
| Equal risk | Options | 0.650 | 1.988 | 175.662 | 0.116 | 0.027 | 0.089 | 0.689 | 0.126 | 0.065 | 0.561 | 0.439 |
| VIX hedge sleeve | Options | 0.000 | 0.000 | 18.435 | 1.572 | 0.000 | 1.572 | -23.137 | -0.406 | 1.000 | 1.000 | 0.000 |
| Delta-matched equities | Equities | 3.307 | 3.307 | 0.000 | 0.000 | 0.000 | 0.000 | | | 0.000 | | |
| Underlying Markowitz | Equities | 0.539 | 1.000 | 0.000 | 0.000 | 0.000 | 0.000 | | | 0.000 | | |

**Table 11:** Greek, beta-proxy, VIX-vega, stress, and option-right exposures.

## *A.2. Option, VIX, cost, and settlement diagnostics*

The option results are reported only after timing, settlement, and cost conventions are visible. This is where the paper records the distinction between gross theory returns, post-cost research returns, and production claims that are not made.

| Diagnostic | Value |
|---|---|
| Return construction | Prior-date option selection, split-adjusted listed-expiry payoff |
| Minimum option mark filter | 0.25 |
| Max train decision date | 2020-11-30 |
| Max train realization date | 2020-12-31 |
| First test decision date | 2020-12-31 |
| First test realization date | 2021-01-29 |
| Detected split/unit adjustments | 7 |
| Max single option return after repair | 48.700 |
| Min single option return after repair | -1.000 |
| Expiry spot source | Raw daily close on listed expiry, or prior trading day if missing |
| VIX option return construction | Prior-date VIX option selection, VX-forward Greeks, expiry settlement proxy flagged |
| VIX settlement source | vro_soq_exact:536 |
| VIX first test decision date | 2020-12-31 |

**Table 12:** Timing and return-construction diagnostics.

| Check | Value | Pass |
|---|---|---|
| OPRA-derived rows after liquidity filters | 160,315 | yes |
| Representative option choices | 5,825 | yes |
| Nonmissing option return cells | 5,777 | yes |
| Max OPRA spot vs raw close mismatch | 2.129e-16 | yes |
| All option choices made before payoff date | True | yes |
| All payoff dates no later than next rebalance | True | yes |
| Exact listed-expiry close share | 1.000 | yes |
| Max expiry-to-payoff-date lag in days | 0 | yes |
| Holding period days min/median/max | 15/18.0/21 | yes |
| Rows requiring split conversion during holding | 13 | yes |
| Detected raw-close split/unit events | 7 | yes |
| Minimum single-option return | -1.000 | yes |
| Maximum single-option return | 48.700 | yes |
| Max training decision before first test decision | 2020-11-30 before 2020-12-31 | yes |
| Raw VIX option rows after contract parsing | 103,650 | yes |
| Representative VIX option choices | 630 | yes |
| VIX option holding ledger rows | 536 | yes |
| Duplicate VIX date-symbol rows after dedupe | 0 | yes |
| Black-76 VX-forward Greek coverage | 1.000 | yes |
| VIX settlement source | vro_soq_exact:536 | yes |
| All VIX decisions before payoff dates | True | yes |
| Minimum VIX long-option return | -1.000 | yes |

**Table 13:** Trading-data audit for the point-in-time holding model. Rows include VIX option settlement-source flags; proxy rows, if present, remain diagnostic.

| Year | Exact scalar parsed | SOQ component status | RequiredExpiries | FirstExpiration | LastExpiration |
|---|---|---|---|---|---|
| 2015 | False | downloaded | 3 | 2015-12-09 | 2015-12-30 |
| 2015 | True | downloaded | 10 | 2015-02-18 | 2015-11-18 |
| 2016 | False | unavailable | 11 | 2016-05-04 | 2016-10-26 |
| 2016 | True | downloaded | 3 | 2016-01-20 | 2016-03-16 |
| 2016 | True | unavailable | 4 | 2016-04-20 | 2016-12-21 |
| 2017 | False | unavailable | 20 | 2017-02-15 | 2017-12-27 |
| 2017 | True | unavailable | 3 | 2017-01-18 | 2017-06-21 |
| 2018 | False | unavailable | 6 | 2018-01-03 | 2018-10-31 |
| 2018 | True | unavailable | 8 | 2018-02-14 | 2018-12-19 |
| 2019 | True | downloaded | 3 | 2019-10-16 | 2019-12-18 |
| 2019 | True | unavailable | 8 | 2019-01-16 | 2019-09-18 |
| 2020 | True | downloaded | 12 | 2020-01-22 | 2020-12-16 |
| 2021 | True | downloaded | 11 | 2021-01-20 | 2021-12-22 |
| 2022 | True | downloaded | 10 | 2022-01-19 | 2022-12-21 |
| 2023 | True | downloaded | 12 | 2023-01-18 | 2023-12-20 |
| 2024 | True | downloaded | 11 | 2024-01-17 | 2024-12-18 |
| 2025 | True | downloaded | 11 | 2025-01-22 | 2025-12-17 |
| 2026 | True | downloaded | 4 | 2026-01-21 | 2026-04-15 |

**Table 14:** Public Cboe VRO/SOQ download audit for required VIX expiries. Rows with `Exact scalar parsed = False` are not used in headline VIX option returns.

| Strategy | Mean monthly cost | Ann. cost drag | Spread share | Fee share | Mean margin/NAV | Mean stress margin/NAV | Mean assignment notional/NAV | Capacity penalized share |
|---|---|---|---|---|---|---|---|---|
| Beta/delta-neutral + VIX | 0.422 | 5.068 | 0.099 | 0.012 | 4.029 | 0.235 | 0.000 | 0.273 |
| Equal premium | 0.259 | 3.109 | 0.115 | 0.025 | 0.953 | 0.238 | 0.000 | 0.381 |
| Equal risk | 0.115 | 1.381 | 0.213 | 0.035 | 0.920 | 0.230 | 0.000 | 0.347 |
| Equity-option Greek Markowitz | 0.733 | 8.794 | 0.021 | 0.003 | 4.610 | 0.224 | 0.039 | 0.335 |
| Greek Markowitz + VIX | 0.610 | 7.325 | 0.069 | 0.008 | 3.526 | 0.234 | 0.000 | 0.290 |
| VIX hedge sleeve | 0.430 | 5.165 | 0.347 | 0.046 | 0.978 | 0.244 | 0.000 | 0.847 |

**Table 15:** Research cost and capital diagnostics. Costs include transaction costs, spread crossing, explicit fees, slippage, borrow, capacity penalties, simulated margin funding, and assignment-risk penalties where flagged. These are not live broker fills.

| Strategy | Scenario | Avg contracts traded | Avg quoted spread | Avg monthly cost | Max capacity used | Rejected trades | Mean margin/NAV | Mean stress capital/NAV |
|---|---|---|---|---|---|---|---|---|
| Beta/delta-neutral + VIX | full_spread | 56.74916809036922 | 0.03509309915904966 | 0.03198045377141552 | 0.9997049345720702 | 867 | 1.6454245692301974 | 0.11046887552661715 |
| Beta/delta-neutral + VIX | half_spread | 56.74916809036922 | 0.03509309915904966 | 0.01874611132937927 | 0.9997049345720702 | 867 | 1.6454245692301974 | 0.11046887552661715 |
| Beta/delta-neutral + VIX | mid | 56.74916809036922 | 0.03509309915904966 | 0.005511768887343023 | 0.9997049345720702 | 867 | 1.6454245692301974 | 0.11046887552661715 |
| Equal premium | full_spread | 65.36655726350517 | 0.03450510899698828 | 0.027393513458707507 | 0.9993803841618195 | 1330 | 0.5895061728395058 | 0.14737654320987645 |
| Equal premium | half_spread | 65.36655726350517 | 0.03450510899698828 | 0.017223026084595215 | 0.9993803841618195 | 1330 | 0.5895061728395058 | 0.14737654320987645 |
| Equal premium | mid | 65.36655726350517 | 0.03450510899698828 | 0.007052538710482932 | 0.9993803841618195 | 1330 | 0.5895061728395058 | 0.14737654320987645 |
| Equal risk | full_spread | 43.65715485388917 | 0.03490292057829765 | 0.02125442824856588 | 0.9990134907215874 | 1222 | 0.5170241047560391 | 0.12925602618900978 |
| Equal risk | half_spread | 43.65715485388917 | 0.03490292057829765 | 0.013262882446192645 | 0.9990134907215874 | 1222 | 0.5170241047560391 | 0.12925602618900978 |
| Equal risk | mid | 43.65715485388917 | 0.03490292057829765 | 0.005271336643819411 | 0.9990134907215874 | 1222 | 0.5170241047560391 | 0.12925602618900978 |
| Equity-option Greek Markowitz | full_spread | 18.797109490588205 | 0.017435798276330978 | 0.006679109598942776 | 0.9901002345178692 | 957 | 1.3950575743846914 | 0.07960287865721945 |
| Equity-option Greek Markowitz | half_spread | 18.797109490588205 | 0.017435798276330978 | 0.0044518190699674685 | 0.9901002345178692 | 957 | 1.3950575743846914 | 0.07960287865721945 |
| Equity-option Greek Markowitz | mid | 18.797109490588205 | 0.017435798276330978 | 0.002224528540992161 | 0.9901002345178692 | 957 | 1.3950575743846914 | 0.07960287865721945 |
| Greek Markowitz + VIX | full_spread | 49.294675424203284 | 0.035957144195152634 | 0.026579996362664688 | 0.9988451122012506 | 898 | 1.5123553557133067 | 0.10149951553572087 |
| Greek Markowitz + VIX | half_spread | 49.294675424203284 | 0.035957144195152634 | 0.01563456341722021 | 0.9988451122012506 | 898 | 1.5123553557133067 | 0.10149951553572087 |
| Greek Markowitz + VIX | mid | 49.294675424203284 | 0.035957144195152634 | 0.0046891304717757345 | 0.9988451122012506 | 898 | 1.5123553557133067 | 0.10149951553572087 |
| VIX hedge sleeve | full_spread | 2410.1807707861826 | 0.15 | 0.07580327078625228 | 0.9885078059495911 | 153 | 0.14999999999999997 | 0.037499999999999999 |
| VIX hedge sleeve | half_spread | 2410.1807707861826 | 0.15 | 0.042053270786252275 | 0.9885078059495911 | 153 | 0.14999999999999997 | 0.037499999999999999 |
| VIX hedge sleeve | mid | 2410.1807707861826 | 0.15 | 0.008303270786252284 | 0.9885078059495911 | 153 | 0.14999999999999997 | 0.037499999999999999 |

**Table 16:** Executable-cost, capacity, and market-impact diagnostics by scenario. Rows are generated from the cost-scenario ledger and record no-fill/capacity failures rather than assuming midpoint execution.

| Strategy | Downside ann. dev | Max drawdown | Worst month | SR 90% CI lo | SR 90% CI hi | Pred./realized vol | Gross NAV | Net NAV |
|---|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | 0.505 | -0.876 | -0.455 | 0.842 | 0.083 | 1.089 | 1.000 | 0.500 |
| Greek Markowitz + VIX | 0.368 | -0.613 | -0.371 | 1.374 | 0.602 | 1.385 | 1.000 | 0.500 |
| Beta/delta-neutral + VIX | 0.344 | -0.570 | -0.354 | 1.414 | 0.676 | 1.478 | 1.000 | 0.500 |
| Equal premium | 1.004 | -1.000 | -0.662 | -0.030 | -0.635 | 2.782 | 1.000 | 1.000 |
| Equal risk | 0.713 | -0.976 | -0.516 | 0.082 | -0.558 | 1.814 | 1.000 | 1.000 |
| VIX hedge sleeve | 3.006 | | -1.000 | -6.429 | -9.004 | 11.150 | 1.000 | 1.000 |
| Delta-matched equities | 0.473 | -0.894 | -0.454 | 1.176 | 0.372 | | 3.307 | 3.307 |
| Underlying Markowitz | 0.096 | -0.303 | -0.073 | 0.909 | 0.103 | | 1.000 | 0.539 |

**Table 17:** Gross performance diagnostics and NAV accounting.

| Strategy | Downside ann. dev | Max drawdown | Worst month | SR 90% CI lo | SR 90% CI hi | Pred./realized vol | Gross NAV | Net NAV |
|---|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | 5.864 | -1.243 | -10.800 | -2.209 | -1.164 | | 1.000 | 0.500 |
| Greek Markowitz + VIX | 5.074 | -1.183 | -9.226 | -1.871 | -0.994 | | 1.000 | 0.500 |
| Beta/delta-neutral + VIX | 2.997 | -1.000 | -5.540 | -1.961 | -1.094 | | 1.000 | 0.500 |
| Equal premium | 2.107 | -1.037 | -2.431 | -1.975 | -0.816 | | 1.000 | 1.000 |
| Equal risk | 1.065 | -1.000 | -0.974 | -1.576 | -0.486 | | 1.000 | 1.000 |
| VIX hedge sleeve | 5.876 | -1.159 | -9.482 | -7.714 | -2.676 | | 1.000 | 1.000 |
| Delta-matched equities | 0.473 | -0.894 | -0.454 | 0.372 | 2.182 | | | |
| Underlying Markowitz | 0.096 | -0.303 | -0.073 | 0.103 | 1.812 | | | |

**Table 18:** Post-cost performance diagnostics and NAV accounting.

## *A.3. Validation, inference, and robustness additional results*

The validation appendix keeps uncertain evidence in view. The bootstrap intervals and leave-one-underlying-out tests are uncertainty screens, not proof that a premium is stable.

| Liquidity tier | Strategy | Ann. return | Ann. vol | Sharpe | Calmar | Omega |
|---|---|---|---|---|---|---|
| all_eligible | Greek Markowitz | 1.2463192285080007 | 0.9068172193093107 | 1.3743885779509968 | 2.033918612174948 | 2.7843631987725814 |
| all_eligible | Equal premium | -0.05552159864526898 | 1.878473564887012 | -0.02955676336526383 | -0.05553271091792252 | 0.9771765473237329 |
| all_eligible | Equal risk | 0.10434553413679803 | 1.2751366252116842 | 0.0818308658646486 | 0.10693488072634062 | 1.0641387065829713 |
| all_eligible | Delta neutral | 1.1720618697652665 | 0.8291774074138495 | 1.4135236431740843 | 2.057353788525215 | 2.947600334500001 |
| all_eligible | VIX hedge sleeve | -9.28491238522107 | 1.444181867492721 | -6.429184989935395 | nan | 0.04042569095025844 |
| tight_spread_quartile | Greek Markowitz | 0.7489053854294916 | 0.9298739055330015 | 0.8053838063132019 | 1.0252766141786784 | 1.9114377746808244 |
| tight_spread_quartile | Equal premium | 1.9315962727008187 | 2.264282965233791 | 0.8530719447873328 | 1.9570493784559104 | 1.9570538593238194 |
| tight_spread_quartile | Equal risk | 1.1814619447081522 | 1.6060721114200187 | 0.7356219788061417 | 1.2213787266592875 | 1.731852284214156 |
| tight_spread_quartile | Delta neutral | 0.6871581895298814 | 0.8626278440164994 | 0.7965870732045813 | 0.9516452083830749 | 1.8586123893126145 |
| top_volume_quartile | Greek Markowitz | 2.0376224599150063 | 1.577048405020137 | 1.2920481409630469 | 2.7131456082828156 | 2.672187767038167 |
| top_volume_quartile | Equal premium | -0.9862571402804206 | 2.1637646209976955 | -0.45580611250851527 | -0.9862571478228792 | 0.710630732496604 |
| top_volume_quartile | Equal risk | -0.9851204806746627 | 1.6783050234078585 | -0.5869734445972999 | -0.9851214694043257 | 0.6612429674757221 |
| top_volume_quartile | Delta neutral | 1.8907359027828463 | 1.4349238479289643 | 1.3176559198676352 | 2.475447246701459 | 2.6359199548389416 |
| top_volume_quartile | VIX hedge sleeve | -9.28491238522107 | 1.444181867492721 | -6.429184989935395 | nan | 0.04042569095025844 |

**Table 19:** Liquidity-tier performance. The tiers test whether the option allocation survives when the universe is restricted to high-volume, tight-spread, high-open-interest, or jointly liquid contracts.

| Strategy | Equity delta | Equity gamma | Equity vega | VIX-forward delta | VIX-forward gamma | VIX-option vega | Theta/carry | VX roll | Skew/tail | Residual | Realized ann. mean | Residual vol share |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | 0.521 | 0.898 | 0.002 | 0.000 | 0.000 | 0.000 | -0.003 | 0.000 | 0.000 | -0.524 | 0.894 | 0.570 |
| Greek Markowitz + VIX | 0.489 | 1.475 | 0.001 | 0.767 | -0.867 | -0.227 | 1.057 | 0.000 | 0.000 | -1.447 | 1.246 | 0.950 |
| Beta/delta-neutral + VIX | 0.168 | 1.486 | 0.001 | 0.554 | -0.745 | -0.190 | 1.034 | 0.000 | 0.000 | -1.135 | 1.172 | 1.069 |
| Equal premium | 0.062 | 7.928 | 0.006 | -0.016 | 0.358 | 0.045 | -1.016 | 0.000 | 0.000 | -7.423 | -0.056 | 0.354 |
| Equal risk | 0.027 | 5.625 | 0.004 | -0.096 | 0.270 | 0.049 | -0.592 | 0.000 | 0.000 | -5.184 | 0.104 | 0.514 |
| VIX hedge sleeve | 0.000 | 0.000 | 0.000 | 2.083 | 2.615 | 0.135 | -12.254 | 0.000 | 0.000 | -1.865 | -9.285 | 4.858 |

**Table 20:** Annualized mean P&L attribution. Equity-option Greeks and VIX-forward Greeks are separated; components reconcile to realized mean after the residual term.

| Strategy | Regime | Months | Ann. return | Ann. vol | Sharpe | Sharpe CI lo | Sharpe CI hi | Sortino | Calmar | Omega | Info. ratio |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | Down | 19 | -0.683 | 0.847 | -0.806 | -1.975 | 0.104 | -1.017 | -0.811 | 0.572 | -0.225 |
| Equity-option Greek Markowitz | Flat | 21 | 0.518 | 0.928 | 0.558 | -0.761 | 1.556 | 1.137 | 0.823 | 1.525 | -0.335 |
| Equity-option Greek Markowitz | Up | 20 | 2.787 | 1.175 | 2.373 | 0.685 | 4.553 | 8.069 | 6.615 | 6.928 | 1.808 |
| Greek Markowitz + VIX | Down | 19 | 0.806 | 0.746 | 1.081 | 0.153 | 1.938 | 2.583 | 1.918 | 2.235 | 1.764 |
| Greek Markowitz + VIX | Flat | 21 | 0.562 | 0.917 | 0.612 | -0.991 | 1.786 | 1.343 | 0.965 | 1.618 | 0.073 |
| Greek Markowitz + VIX | Up | 20 | 2.384 | 0.985 | 2.421 | 0.831 | 4.232 | 6.598 | 7.182 | 5.575 | 1.737 |
| Beta/delta-neutral + VIX | Down | 19 | 1.487 | 0.721 | 2.061 | 1.366 | 3.248 | 4.939 | 5.123 | 4.422 | 2.700 |
| Beta/delta-neutral + VIX | Flat | 21 | 0.649 | 0.815 | 0.796 | -0.666 | 1.860 | 2.154 | 1.757 | 1.990 | 0.151 |
| Beta/delta-neutral + VIX | Up | 20 | 1.423 | 0.951 | 1.496 | -0.182 | 3.218 | 3.419 | 3.357 | 3.019 | 0.777 |
| Equal premium | Down | 19 | -0.515 | 1.890 | -0.273 | -1.562 | 0.450 | -0.473 | -0.535 | 0.812 | -0.013 |
| Equal premium | Flat | 21 | -2.368 | 0.879 | -2.694 | -5.258 | -1.256 | -2.270 | -2.377 | 0.155 | -3.484 |
| Equal premium | Up | 20 | 2.809 | 2.352 | 1.195 | 0.314 | 2.010 | 3.236 | 3.712 | 2.597 | 0.917 |
| Equal risk | Down | 19 | 0.101 | 1.436 | 0.070 | -0.954 | 0.868 | 0.128 | 0.130 | 1.054 | 0.411 |
| Equal risk | Flat | 21 | -1.654 | 0.687 | -2.409 | -4.471 | -0.990 | -2.113 | -1.703 | 0.155 | -3.447 |
| Equal risk | Up | 20 | 1.954 | 1.419 | 1.377 | 0.302 | 2.376 | 3.665 | 3.495 | 2.843 | 0.915 |
| VIX hedge sleeve | Down | 19 | -7.743 | 1.368 | -5.662 | -9.800 | -4.100 | -2.982 | | 0.021 | -5.203 |
| VIX hedge sleeve | Flat | 21 | -8.367 | 1.909 | -4.383 | -6.184 | -3.226 | -2.829 | | 0.104 | -4.129 |
| VIX hedge sleeve | Up | 20 | -11.714 | 0.276 | -42.492 | -107.213 | -26.949 | -3.453 | | 0.000 | -40.570 |
| Delta-matched equities | Down | 19 | -1.767 | 0.675 | -2.619 | -4.396 | -1.375 | -2.204 | -1.817 | 0.159 | -2.086 |
| Delta-matched equities | Flat | 21 | 1.272 | 0.707 | 1.800 | 0.747 | 2.788 | 5.921 | 3.839 | 5.091 | 1.172 |
| Delta-matched equities | Up | 20 | 3.630 | 0.698 | 5.203 | 3.511 | 8.613 | 34.716 | 26.891 | 45.819 | 4.428 |
| Underlying Markowitz | Down | 19 | -0.319 | 0.120 | -2.664 | -4.820 | -1.265 | -2.262 | -0.765 | 0.148 | 1.948 |
| Underlying Markowitz | Flat | 21 | 0.172 | 0.185 | 0.933 | -0.496 | 2.168 | 2.087 | 1.120 | 2.038 | -0.742 |
| Underlying Markowitz | Up | 20 | 0.658 | 0.172 | 3.829 | 2.463 | 6.121 | 15.942 | 17.455 | 15.149 | -0.128 |

**Table 21:** Performance by SPY-return tercile. Down, flat, and up regimes are descriptive cuts of the held-out sample, not fitted optimizer inputs.

| Ablation | Ann. return | Ann. vol | Sharpe | Calmar | Omega | Active assets |
|---|---|---|---|---|---|---|
| carry_only | 6.70147099899699 | 1.9164807271628144 | 3.496758878925928 | 2.8096258939800632e-05 | 8.037348345976534 | 54 |
| variance_risk_premium_only | 9.63139175996257 | 0.6700827342459895 | 14.373436693306674 | nan | nan | 54 |
| skew_tail_only | 4.489045206711676 | 3.2048741838250314 | 1.4006931159319276 | 0.003079353839745213 | 2.691677332065947 | 18 |
| vix_regime_only | 4.590815059335688 | 2.544245648201727 | 1.8043914362516356 | 0.004298213073203248 | 3.3862924961758027 | 54 |
| relative_value_only | 2.5789118807598546 | 1.4076551542781681 | 1.8320622582327668 | 7.262838528939612 | 6.69566248471262 | 54 |
| full_conditional_model | 4.113956564258069 | 2.9681846972242805 | 1.3860177124776856 | 2.1877214234861713 | 3.1853840590190665 | 54 |

**Table 22:** Forecast ablation performance. Each row isolates one economically identified component of the conditional option-premium forecast.

| Strategy | Metric | Estimate | CI lo | CI hi | N | Method | Block size | Seed | Return basis |
|---|---|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | Sharpe | 0.842 | 0.083 | 1.679 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equity-option Greek Markowitz | Sortino | 1.770 | 0.141 | 4.581 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equity-option Greek Markowitz | Calmar | 1.020 | 0.083 | 2.828 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equity-option Greek Markowitz | Omega | 1.888 | 1.064 | 3.546 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equity-option Greek Markowitz | Information Ratio | 0.693 | | | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Greek Markowitz + VIX | Sharpe | 1.374 | 0.602 | 2.222 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Greek Markowitz + VIX | Sortino | 3.385 | 1.188 | 7.275 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Greek Markowitz + VIX | Calmar | 2.034 | 0.700 | 4.901 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Greek Markowitz + VIX | Omega | 2.784 | 1.562 | 5.251 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Greek Markowitz + VIX | Information Ratio | 1.192 | | | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Beta/delta-neutral + VIX | Sharpe | 1.414 | 0.676 | 2.220 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Beta/delta-neutral + VIX | Sortino | 3.410 | 1.335 | 6.541 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Beta/delta-neutral + VIX | Calmar | 2.057 | 0.701 | 5.611 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Beta/delta-neutral + VIX | Omega | 2.948 | 1.693 | 5.333 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Beta/delta-neutral + VIX | Information Ratio | 1.219 | | | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equal premium | Sharpe | -0.030 | -0.635 | 0.437 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equal premium | Sortino | -0.055 | -0.946 | 0.920 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equal premium | Calmar | -0.056 | -1.051 | 0.849 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equal premium | Omega | 0.977 | 0.624 | 1.416 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equal premium | Information Ratio | -0.260 | | | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equal risk | Sharpe | 0.082 | -0.558 | 0.627 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equal risk | Sortino | 0.146 | -0.821 | 1.305 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equal risk | Calmar | 0.107 | -0.667 | 0.954 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equal risk | Omega | 1.064 | 0.662 | 1.621 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equal risk | Information Ratio | -0.151 | | | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| VIX hedge sleeve | Sharpe | -6.429 | -9.004 | -4.921 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| VIX hedge sleeve | Sortino | -3.089 | -3.253 | -2.904 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| VIX hedge sleeve | Calmar | | -10.058 | -8.262 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| VIX hedge sleeve | Omega | 0.040 | 0.016 | 0.071 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| VIX hedge sleeve | Information Ratio | -6.344 | | | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Delta-matched equities | Sharpe | 1.176 | 0.372 | 2.182 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Delta-matched equities | Sortino | 2.317 | 0.615 | 5.886 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Delta-matched equities | Calmar | 1.225 | 0.388 | 3.716 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Delta-matched equities | Omega | 2.367 | 1.301 | 5.182 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Delta-matched equities | Information Ratio | 0.995 | | | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Underlying Markowitz | Sharpe | 0.909 | 0.103 | 1.812 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Underlying Markowitz | Sortino | 1.856 | 0.172 | 4.958 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Underlying Markowitz | Calmar | 0.590 | 0.045 | 2.544 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Underlying Markowitz | Omega | 1.929 | 1.074 | 3.877 | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Underlying Markowitz | Information Ratio | 0.166 | | | 60 | circular block bootstrap | 8 | 20260625 | Gross before costs |
| Equity-option Greek Markowitz | Sharpe | -1.443 | -2.209 | -1.164 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equity-option Greek Markowitz | Sortino | -1.347 | -1.895 | -1.123 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equity-option Greek Markowitz | Calmar | -6.357 | -11.009 | -1.856 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equity-option Greek Markowitz | Omega | 0.098 | 0.045 | 0.218 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equity-option Greek Markowitz | Information Ratio | -1.492 | | | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Greek Markowitz + VIX | Sharpe | -1.259 | -1.871 | -0.994 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Greek Markowitz + VIX | Sortino | -1.198 | -1.683 | -0.971 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Greek Markowitz + VIX | Calmar | -5.136 | -8.117 | -1.087 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Greek Markowitz + VIX | Omega | 0.116 | 0.054 | 0.269 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Greek Markowitz + VIX | Information Ratio | -1.303 | | | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Beta/delta-neutral + VIX | Sharpe | -1.371 | -1.961 | -1.094 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Beta/delta-neutral + VIX | Sortino | -1.300 | -1.790 | -1.073 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Beta/delta-neutral + VIX | Calmar | -3.895 | -6.033 | -1.289 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Beta/delta-neutral + VIX | Omega | 0.151 | 0.075 | 0.321 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Beta/delta-neutral + VIX | Information Ratio | -1.418 | | | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equal premium | Sharpe | -1.331 | -1.975 | -0.816 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equal premium | Sortino | -1.502 | -1.923 | -1.066 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equal premium | Calmar | -3.051 | -4.189 | -1.543 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equal premium | Omega | 0.359 | 0.206 | 0.547 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equal premium | Information Ratio | -1.619 | | | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equal risk | Sharpe | -0.934 | -1.576 | -0.486 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equal risk | Sortino | -1.199 | -1.714 | -0.697 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equal risk | Calmar | -1.276 | -1.962 | -0.685 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equal risk | Omega | 0.505 | 0.326 | 0.702 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Equal risk | Information Ratio | -1.225 | | | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| VIX hedge sleeve | Sharpe | -3.413 | -7.714 | -2.676 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| VIX hedge sleeve | Sortino | -2.445 | -3.171 | -2.131 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| VIX hedge sleeve | Calmar | -12.391 | -15.461 | -7.145 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| VIX hedge sleeve | Omega | 0.016 | 0.004 | 0.032 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| VIX hedge sleeve | Information Ratio | -3.402 | | | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Delta-matched equities | Sharpe | 1.176 | 0.372 | 2.182 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Delta-matched equities | Sortino | 2.317 | 0.615 | 5.886 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Delta-matched equities | Calmar | 1.225 | 0.388 | 3.716 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Delta-matched equities | Omega | 2.367 | 1.301 | 5.182 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Delta-matched equities | Information Ratio | 0.995 | | | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Underlying Markowitz | Sharpe | 0.909 | 0.103 | 1.812 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Underlying Markowitz | Sortino | 1.856 | 0.172 | 4.958 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Underlying Markowitz | Calmar | 0.590 | 0.045 | 2.544 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Underlying Markowitz | Omega | 1.929 | 1.074 | 3.877 | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |
| Underlying Markowitz | Information Ratio | 0.166 | | | 60 | circular block bootstrap | 8 | 20260625 | Post-cost research |

**Table 23:** Full bootstrap inference ledger for gross and post-cost strategy metrics.

| Variant | Sharpe | Probabilistic Sharpe | Deflated Sharpe | Bootstrap max Sharpe p95 | N | Block size | Seed |
|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz::gross | 0.8421194565895301 | 0.9999999999997874 | 0.0 | 2.395856192979967 | 60 | 8 | 20260625 |
| Greek Markowitz + VIX::gross | 1.3743885779509968 | 1.0 | 1.171660871221686e-190 | 2.395856192979967 | 60 | 8 | 20260625 |
| Beta/delta-neutral + VIX::gross | 1.4135236431740843 | 1.0 | 5.5953905332384846e-164 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equal premium::gross | -0.02955676336526383 | 0.4122235847317047 | 0.0 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equal risk::gross | 0.0818308658646486 | 0.7451516479581755 | 0.0 | 2.395856192979967 | 60 | 8 | 20260625 |
| VIX hedge sleeve::gross | -6.429184989935395 | 1.1855878159463198e-08 | 9.87288524527083e-25 | 2.395856192979967 | 60 | 8 | 20260625 |
| Delta-matched equities::gross | 1.1763702582317712 | 0.9999999999999847 | 7.411543205687691e-158 | 2.395856192979967 | 60 | 8 | 20260625 |
| Underlying Markowitz::gross | 0.9088833301008382 | 0.9999999999999218 | 3.601643297065905e-281 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equity-option Greek Markowitz::mid | 0.8170505717670248 | 0.9999999999992117 | 0.0 | 2.395856192979967 | 60 | 8 | 20260625 |
| Greek Markowitz + VIX::mid | 1.3138775407676593 | 1.0 | 2.0832065107842354e-203 | 2.395856192979967 | 60 | 8 | 20260625 |
| Beta/delta-neutral + VIX::mid | 1.3352567190175022 | 1.0 | 8.293236346779767e-180 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equal premium::mid | -0.07459766125108384 | 0.29440021003850236 | 0.0 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equal risk::mid | 0.032216351568761376 | 0.5995571041200423 | 0.0 | 2.395856192979967 | 60 | 8 | 20260625 |
| VIX hedge sleeve::mid | -6.4606628766839735 | 1.0514523448932735e-08 | 8.416119683516885e-25 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equity-option Greek Markowitz::half_spread | 0.7919362861614629 | 0.9999999999969282 | 0.0 | 2.395856192979967 | 60 | 8 | 20260625 |
| Greek Markowitz + VIX::half_spread | 1.1703215522353334 | 1.0 | 2.6433421736216467e-233 | 2.395856192979967 | 60 | 8 | 20260625 |
| Beta/delta-neutral + VIX::half_spread | 1.143232107598715 | 1.0 | 4.726688767518834e-220 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equal premium::half_spread | -0.13959427042490616 | 0.16788800945156535 | 2.979791317633e-310 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equal risk::half_spread | -0.042972181890083776 | 0.37381543239837334 | 0.0 | 2.395856192979967 | 60 | 8 | 20260625 |
| VIX hedge sleeve::half_spread | -6.540853900526196 | 7.56882120672634e-09 | 5.219977061164737e-25 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equity-option Greek Markowitz::full_spread | 0.7668067218645858 | 0.9999999999880843 | 0.0 | 2.395856192979967 | 60 | 8 | 20260625 |
| Greek Markowitz + VIX::full_spread | 1.0258219183556512 | 0.9999999999999999 | 4.061857790130324e-263 | 2.395856192979967 | 60 | 8 | 20260625 |
| Beta/delta-neutral + VIX::full_spread | 0.9508576292459209 | 0.9999999999999716 | 2.0390393622011163e-261 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equal premium::full_spread | -0.20462197552298175 | 0.08945368147995542 | 1.711455237880309e-288 | 2.395856192979967 | 60 | 8 | 20260625 |
| Equal risk::full_spread | -0.11815498556113241 | 0.19879777872024146 | 0.0 | 2.395856192979967 | 60 | 8 | 20260625 |
| VIX hedge sleeve::full_spread | -6.607601788466617 | 4.860837695911706e-09 | 2.0508103641611321e-25 | 2.395856192979967 | 60 | 8 | 20260625 |

**Table 24:** Reality-check inference across gross and executable-cost variants.

### A.3.1 Tail-path simulation diagnostics

The next tables simulate alternative out-of-sample paths from the realized monthly strategy returns. They are path-risk diagnostics only: simulated returns do not enter contract selection, optimizer weights, expected-return forecasts, or any trading decision. The block bootstrap preserves short-run dependence in the realized option return path. The volatility-clustered simulation fits EGARCH(1,1)-t only when the OOS sample is long enough; otherwise it reports an EWMA residual fallback rather than pretending that the EGARCH parameters are precisely estimated.

| Return basis | Strategy | Requested method | Simulation | Breach 10% | Breach 25% | Breach 50% | Breach 75% | Breach 90% |
|---|---|---|---|---|---|---|---|---|
| Gross before costs | Equity-option Greek Markowitz | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 1.000 | 0.983 | 0.770 | 0.298 |
| Gross before costs | Equity-option Greek Markowitz | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 1.000 | 0.987 | 0.708 | 0.296 |
| Gross before costs | Greek Markowitz + VIX | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 0.996 | 0.795 | 0.135 | 0.005 |
| Gross before costs | Greek Markowitz + VIX | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 0.999 | 0.884 | 0.379 | 0.136 |
| Gross before costs | Beta/delta-neutral + VIX | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 0.996 | 0.585 | 0.063 | 0.001 |
| Gross before costs | Beta/delta-neutral + VIX | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 0.996 | 0.671 | 0.176 | 0.037 |
| Gross before costs | Delta-matched equities | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 1.000 | 0.902 | 0.548 | 0.175 |
| Gross before costs | Delta-matched equities | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 1.000 | 0.896 | 0.358 | 0.069 |
| Gross before costs | Underlying Markowitz | circular_block_bootstrap | circular_block_bootstrap | 0.981 | 0.432 | 0.013 | 0.000 | 0.000 |
| Gross before costs | Underlying Markowitz | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 0.981 | 0.272 | 0.002 | 0.000 | 0.000 |
| Gross before costs | Equal premium | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 1.000 | 1.000 | 0.999 | 0.995 |
| Gross before costs | Equal premium | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 0.986 | 0.986 | 0.986 | 0.986 | 0.985 |
| Gross before costs | Equal risk | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 1.000 | 1.000 | 0.973 | 0.841 |
| Gross before costs | Equal risk | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 1.000 | 1.000 | 0.993 | 0.937 |
| Gross before costs | VIX hedge sleeve | circular_block_bootstrap | circular_block_bootstrap | 0.437 | 0.437 | 0.437 | 0.437 | 0.437 |
| Gross before costs | VIX hedge sleeve | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Full-spread post-cost | Equity-option Greek Markowitz | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 1.000 | 0.986 | 0.789 | 0.343 |
| Full-spread post-cost | Equity-option Greek Markowitz | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 1.000 | 0.993 | 0.756 | 0.359 |
| Full-spread post-cost | Greek Markowitz + VIX | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 0.997 | 0.913 | 0.374 | 0.060 |
| Full-spread post-cost | Greek Markowitz + VIX | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 1.000 | 0.948 | 0.503 | 0.181 |
| Full-spread post-cost | Beta/delta-neutral + VIX | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 1.000 | 0.799 | 0.251 | 0.035 |
| Full-spread post-cost | Beta/delta-neutral + VIX | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 1.000 | 0.850 | 0.295 | 0.068 |
| Full-spread post-cost | Delta-matched equities | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 1.000 | 0.902 | 0.548 | 0.175 |
| Full-spread post-cost | Delta-matched equities | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 1.000 | 0.896 | 0.358 | 0.069 |
| Full-spread post-cost | Underlying Markowitz | circular_block_bootstrap | circular_block_bootstrap | 0.981 | 0.432 | 0.013 | 0.000 | 0.000 |
| Full-spread post-cost | Underlying Markowitz | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 0.981 | 0.272 | 0.002 | 0.000 | 0.000 |
| Full-spread post-cost | Equal premium | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 1.000 | 1.000 | 1.000 | 0.999 |
| Full-spread post-cost | Equal premium | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 0.986 | 0.986 | 0.986 | 0.986 | 0.986 |
| Full-spread post-cost | Equal risk | circular_block_bootstrap | circular_block_bootstrap | 1.000 | 1.000 | 1.000 | 0.989 | 0.938 |
| Full-spread post-cost | Equal risk | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 1.000 | 1.000 | 1.000 | 0.984 |
| Full-spread post-cost | VIX hedge sleeve | circular_block_bootstrap | circular_block_bootstrap | 0.471 | 0.471 | 0.471 | 0.471 | 0.471 |
| Full-spread post-cost | VIX hedge sleeve | egarch_or_ewma | ewma_residual_fallback_insufficient_egarch_obs | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |

**Table 25:** Drawdown-breach probabilities from block-bootstrap and volatility-clustered path simulations. The thresholds are intentionally wide because option-only post-cost paths can experience large drawdowns.

| Return basis | Strategy | Simulation | N paths | Ann. return p50 | Sortino p50 | Max DD p05 | Max DD p50 | Terminal wealth p05 |
|---|---|---|---|---|---|---|---|---|
| Gross before costs | Equity-option Greek Markowitz | circular_block_bootstrap | 1000 | 0.525 | 1.826 | -0.976 | -0.840 | 0.128 |
| Gross before costs | Equity-option Greek Markowitz | ewma_residual_fallback_insufficient_egarch_obs | 1000 | 0.376 | 1.561 | -0.984 | -0.835 | 0.125 |
| Gross before costs | Greek Markowitz + VIX | circular_block_bootstrap | 1000 | 1.382 | 3.451 | -0.816 | -0.602 | 3.033 |
| Gross before costs | Greek Markowitz + VIX | ewma_residual_fallback_insufficient_egarch_obs | 1000 | 1.485 | 3.383 | -0.981 | -0.700 | 1.391 |
| Gross before costs | Beta/delta-neutral + VIX | circular_block_bootstrap | 1000 | 1.354 | 3.505 | -0.777 | -0.547 | 4.583 |
| Gross before costs | Beta/delta-neutral + VIX | ewma_residual_fallback_insufficient_egarch_obs | 1000 | 1.274 | 3.238 | -0.878 | -0.577 | 3.275 |
| Gross before costs | Delta-matched equities | circular_block_bootstrap | 1000 | 1.006 | 2.410 | -0.964 | -0.774 | 0.717 |
| Gross before costs | Delta-matched equities | ewma_residual_fallback_insufficient_egarch_obs | 1000 | 1.283 | 2.834 | -0.919 | -0.696 | 1.819 |
| Gross before costs | Underlying Markowitz | circular_block_bootstrap | 1000 | 0.176 | 1.943 | -0.422 | -0.232 | 1.035 |
| Gross before costs | Underlying Markowitz | ewma_residual_fallback_insufficient_egarch_obs | 1000 | 0.149 | 1.691 | -0.352 | -0.199 | 1.036 |
| Gross before costs | Equal premium | circular_block_bootstrap | 1000 | -0.784 | -0.064 | -1.000 | -1.000 | 0.000 |
| Gross before costs | Equal premium | ewma_residual_fallback_insufficient_egarch_obs | 1000 | -0.856 | -0.612 | -1.008 | -1.000 | -0.000 |
| Gross before costs | Equal risk | circular_block_bootstrap | 1000 | -0.450 | 0.153 | -0.999 | -0.982 | 0.001 |
| Gross before costs | Equal risk | ewma_residual_fallback_insufficient_egarch_obs | 1000 | -0.590 | -0.340 | -1.000 | -0.995 | 0.000 |
| Gross before costs | VIX hedge sleeve | circular_block_bootstrap | 1000 | -1.000 | -3.084 | -1.000 | -1.000 | 0.000 |
| Gross before costs | VIX hedge sleeve | ewma_residual_fallback_insufficient_egarch_obs | 1000 | -1.000 | -1.180 | -5119394349261.397 | -26003204.190 | -22351740928646619564915026613239192969479982290743869848025283945750791128187332694099412714196597845910582341198727462815578867943076576915568644054240573677577398675852626329317101896386114769387520000.000 |
| Full-spread post-cost | Equity-option Greek Markowitz | circular_block_bootstrap | 1000 | 0.413 | 1.626 | -0.982 | -0.856 | 0.083 |
| Full-spread post-cost | Equity-option Greek Markowitz | ewma_residual_fallback_insufficient_egarch_obs | 1000 | 0.236 | 1.335 | -0.990 | -0.857 | 0.068 |
| Full-spread post-cost | Greek Markowitz + VIX | circular_block_bootstrap | 1000 | 0.761 | 2.263 | -0.908 | -0.696 | 0.615 |
| Full-spread post-cost | Greek Markowitz + VIX | ewma_residual_fallback_insufficient_egarch_obs | 1000 | 0.727 | 2.177 | -0.977 | -0.751 | 0.324 |
| Full-spread post-cost | Beta/delta-neutral + VIX | circular_block_bootstrap | 1000 | 0.632 | 2.038 | -0.888 | -0.644 | 0.661 |
| Full-spread post-cost | Beta/delta-neutral + VIX | ewma_residual_fallback_insufficient_egarch_obs | 1000 | 0.509 | 1.792 | -0.918 | -0.653 | 0.519 |
| Full-spread post-cost | Delta-matched equities | circular_block_bootstrap | 1000 | 1.006 | 2.410 | -0.964 | -0.774 | 0.717 |
| Full-spread post-cost | Delta-matched equities | ewma_residual_fallback_insufficient_egarch_obs | 1000 | 1.283 | 2.834 | -0.919 | -0.696 | 1.819 |
| Full-spread post-cost | Underlying Markowitz | circular_block_bootstrap | 1000 | 0.176 | 1.943 | -0.422 | -0.232 | 1.035 |
| Full-spread post-cost | Underlying Markowitz | ewma_residual_fallback_insufficient_egarch_obs | 1000 | 0.149 | 1.691 | -0.352 | -0.199 | 1.036 |
| Full-spread post-cost | Equal premium | circular_block_bootstrap | 1000 | -0.858 | -0.370 | -1.000 | -1.000 | 0.000 |
| Full-spread post-cost | Equal premium | ewma_residual_fallback_insufficient_egarch_obs | 1000 | -0.921 | -0.879 | -1.029 | -1.000 | -0.000 |
| Full-spread post-cost | Equal risk | circular_block_bootstrap | 1000 | -0.589 | -0.195 | -1.000 | -0.994 | 0.000 |
| Full-spread post-cost | Equal risk | ewma_residual_fallback_insufficient_egarch_obs | 1000 | -0.718 | -0.665 | -1.000 | -0.999 | 0.000 |
| Full-spread post-cost | VIX hedge sleeve | circular_block_bootstrap | 1000 | -1.000 | -3.097 | -1.072 | -1.000 | 0.000 |
| Full-spread post-cost | VIX hedge sleeve | ewma_residual_fallback_insufficient_egarch_obs | 1000 | -1.000 | -1.168 | -14999937378889.883 | -52339900.203 | -525374998467292025864430357839332563654903718092683760000481729782174345503400605505835247062471512248274435793844560577271773102709211796292106095746096284591255375346799472801124129161225082342098369467383808.000 |

**Table 26:** Tail-path simulation summary. Median return and Sortino describe the simulated path center; the fifth-percentile maximum drawdown and terminal wealth columns describe the left tail of realized-path resampling.

| Return basis | Strategy | Method | Status | Reason | N obs | Source start | Source end | Block length | Path count | Lag1 autocorr |
|---|---|---|---|---|---|---|---|---|---|---|
| Gross before costs | Equity-option Greek Markowitz | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | 0.096 |
| Gross before costs | Equity-option Greek Markowitz | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | 0.096 |
| Gross before costs | Greek Markowitz + VIX | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | 0.003 |
| Gross before costs | Greek Markowitz + VIX | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | 0.003 |
| Gross before costs | Beta/delta-neutral + VIX | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | -0.083 |
| Gross before costs | Beta/delta-neutral + VIX | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | -0.083 |
| Gross before costs | Delta-matched equities | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | 0.014 |
| Gross before costs | Delta-matched equities | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | 0.014 |
| Gross before costs | Underlying Markowitz | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | 0.041 |
| Gross before costs | Underlying Markowitz | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | 0.041 |
| Gross before costs | Equal premium | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | -0.256 |
| Gross before costs | Equal premium | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | -0.256 |
| Gross before costs | Equal risk | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | -0.188 |
| Gross before costs | Equal risk | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | -0.188 |
| Gross before costs | VIX hedge sleeve | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | -0.125 |
| Gross before costs | VIX hedge sleeve | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | -0.125 |
| Full-spread post-cost | Equity-option Greek Markowitz | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | 0.096 |
| Full-spread post-cost | Equity-option Greek Markowitz | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | 0.096 |
| Full-spread post-cost | Greek Markowitz + VIX | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | -0.008 |
| Full-spread post-cost | Greek Markowitz + VIX | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | -0.008 |
| Full-spread post-cost | Beta/delta-neutral + VIX | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | -0.096 |
| Full-spread post-cost | Beta/delta-neutral + VIX | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | -0.096 |
| Full-spread post-cost | Delta-matched equities | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | 0.014 |
| Full-spread post-cost | Delta-matched equities | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | 0.014 |
| Full-spread post-cost | Underlying Markowitz | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | 0.041 |
| Full-spread post-cost | Underlying Markowitz | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | 0.041 |
| Full-spread post-cost | Equal premium | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | -0.256 |
| Full-spread post-cost | Equal premium | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | -0.256 |
| Full-spread post-cost | Equal risk | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | -0.188 |
| Full-spread post-cost | Equal risk | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | -0.188 |
| Full-spread post-cost | VIX hedge sleeve | circular_block_bootstrap | ok | nan | 60 | 2021-01-29 | 2026-04-30 | 6.000 | 1000 | -0.107 |
| Full-spread post-cost | VIX hedge sleeve | ewma_residual_fallback_insufficient_egarch_obs | fallback | insufficient_egarch_obs | 60 | 2021-01-29 | 2026-04-30 | | 1000 | -0.107 |

**Table 27:** Simulation assumptions and model status. Fallback rows are deliberate diagnostics: when the monthly OOS sample is too short for EGARCH estimation, the appendix uses an EWMA residual bootstrap and records the reason.

| Strategy | Regime | Months | Ann. return | Ann. vol | Sharpe | Sharpe CI lo | Sharpe CI hi | Sortino | Calmar | Omega | Info. ratio |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | Low VIX | 20 | 1.585 | 0.919 | 1.724 | 1.033 | 2.559 | 4.862 | 5.911 | 3.697 | 1.954 |
| Equity-option Greek Markowitz | Mid VIX | 20 | 1.104 | 1.216 | 0.907 | -0.388 | 1.845 | 2.157 | 2.351 | 2.017 | 0.841 |
| Equity-option Greek Markowitz | High VIX | 20 | -0.006 | 1.029 | -0.006 | -1.030 | 1.097 | -0.009 | -0.007 | 0.996 | -0.808 |
| Greek Markowitz + VIX | Low VIX | 20 | 1.386 | 0.846 | 1.638 | 0.265 | 3.472 | 4.774 | 4.419 | 3.257 | 2.330 |
| Greek Markowitz + VIX | Mid VIX | 20 | 1.533 | 1.054 | 1.455 | 0.412 | 2.331 | 3.812 | 4.985 | 3.198 | 1.408 |
| Greek Markowitz + VIX | High VIX | 20 | 0.821 | 0.838 | 0.979 | -0.096 | 2.111 | 2.047 | 2.057 | 2.047 | -0.223 |
| Beta/delta-neutral + VIX | Low VIX | 20 | 1.176 | 0.761 | 1.546 | 0.252 | 3.145 | 3.811 | 5.194 | 2.948 | 2.396 |
| Beta/delta-neutral + VIX | Mid VIX | 20 | 0.974 | 0.965 | 1.009 | -0.446 | 2.152 | 2.326 | 2.350 | 2.250 | 1.028 |
| Beta/delta-neutral + VIX | High VIX | 20 | 1.366 | 0.786 | 1.737 | 0.804 | 2.874 | 4.716 | 4.063 | 4.231 | 0.383 |
| Equal premium | Low VIX | 20 | 0.249 | 1.592 | 0.157 | -1.511 | 1.123 | 0.283 | 0.286 | 1.132 | 0.551 |
| Equal premium | Mid VIX | 20 | 0.944 | 2.287 | 0.413 | -0.680 | 1.136 | 0.923 | 1.004 | 1.371 | 0.398 |
| Equal premium | High VIX | 20 | -1.360 | 1.717 | -0.792 | -3.832 | 0.263 | -1.240 | -1.372 | 0.524 | -1.345 |
| Equal risk | Low VIX | 20 | 0.598 | 1.122 | 0.533 | -0.926 | 1.641 | 0.983 | 0.824 | 1.488 | 1.038 |
| Equal risk | Mid VIX | 20 | 0.348 | 1.456 | 0.239 | -0.779 | 0.996 | 0.488 | 0.436 | 1.204 | 0.231 |
| Equal risk | High VIX | 20 | -0.633 | 1.264 | -0.501 | -2.543 | 0.500 | -0.787 | -0.679 | 0.676 | -1.227 |
| VIX hedge sleeve | Low VIX | 20 | -9.290 | 1.621 | -5.730 | -20.577 | -3.438 | -3.036 | -9.290 | 0.063 | -5.189 |
| VIX hedge sleeve | Mid VIX | 20 | -10.031 | 1.331 | -7.534 | -41.311 | -4.785 | -3.203 | | 0.038 | -7.879 |
| VIX hedge sleeve | High VIX | 20 | -8.533 | 1.406 | -6.069 | -9.076 | -4.610 | -3.031 | | 0.018 | -7.261 |
| Delta-matched equities | Low VIX | 20 | 2.273 | 0.692 | 3.283 | 2.654 | 4.097 | 19.209 | 15.977 | 17.371 | 2.903 |
| Delta-matched equities | Mid VIX | 20 | 1.918 | 0.917 | 2.091 | 1.374 | 2.983 | 6.416 | 6.081 | 5.040 | 1.689 |
| Delta-matched equities | High VIX | 20 | -0.904 | 0.917 | -0.986 | -2.049 | -0.174 | -1.201 | -0.980 | 0.495 | -1.403 |
| Underlying Markowitz | Low VIX | 20 | 0.350 | 0.174 | 2.008 | 0.968 | 3.220 | 5.384 | 4.524 | 4.537 | 1.879 |
| Underlying Markowitz | Mid VIX | 20 | 0.386 | 0.194 | 1.994 | 1.369 | 2.695 | 7.002 | 7.409 | 5.008 | 0.746 |
| Underlying Markowitz | High VIX | 20 | -0.201 | 0.181 | -1.108 | -2.171 | -0.324 | -1.403 | -0.622 | 0.473 | -1.495 |

**Table 28:** Performance by VIX-level tercile.

| Strategy | Ann. alpha | Alpha HAC t | Alpha HAC se | HAC lags | $R^2$ | Residual ann. vol | N | Beta SPY | Beta AAPL | Beta AMZN | Beta GOOGL | Beta JPM | Beta META | Beta MSFT | Beta NVDA | Beta TSLA | Beta VX front | Beta dVIX | Beta dVVIX |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Equity-option Greek Markowitz | 0.405 | 1.141 | 0.355 | 3 | 0.618 | 0.634 | 58 | 0.005 | -1.331 | -0.875 | 0.631 | 0.590 | 0.126 | 2.585 | -0.089 | 1.136 | -1.368 | 0.046 | 0.003 |
| Greek Markowitz + VIX | 0.675 | 2.060 | 0.328 | 3 | 0.513 | 0.616 | 58 | 1.908 | -1.462 | -0.747 | 0.625 | -0.127 | -0.026 | 1.877 | 0.222 | 0.871 | -0.767 | 0.056 | -0.001 |
| Beta/delta-neutral + VIX | 0.845 | 2.753 | 0.307 | 3 | 0.429 | 0.601 | 58 | 0.715 | -1.475 | -0.612 | 0.472 | -0.579 | -0.170 | 1.336 | 0.370 | 0.815 | -0.522 | 0.038 | -0.001 |
| Equal premium | -0.416 | -0.566 | 0.735 | 3 | 0.253 | 1.647 | 58 | 2.522 | -1.714 | -0.069 | 1.943 | -1.879 | -0.344 | 3.001 | -0.374 | 0.404 | -0.749 | 0.032 | -0.001 |
| Equal risk | -0.183 | -0.334 | 0.547 | 3 | 0.253 | 1.115 | 58 | 2.041 | -1.795 | -0.113 | 1.331 | -1.484 | -0.316 | 1.968 | -0.082 | 0.321 | -0.379 | 0.020 | -0.003 |
| VIX hedge sleeve | -8.807 | -14.623 | 0.602 | 3 | 0.252 | 1.264 | 58 | 3.562 | -0.597 | -0.589 | 0.670 | -3.047 | 0.423 | -0.958 | -0.030 | -0.131 | -1.180 | 0.093 | -0.013 |
| Delta-matched equities | 0.000 | 2.229 | 0.000 | 3 | 1.000 | 0.000 | 58 | -0.000 | 0.088 | 0.539 | 0.230 | 0.590 | 0.011 | 0.442 | 0.779 | 0.628 | -0.000 | -0.000 | 0.000 |
| Underlying Markowitz | 0.000 | 2.700 | 0.000 | 3 | 1.000 | 0.000 | 58 | -0.000 | -0.085 | 0.159 | -0.096 | -0.050 | 0.029 | 0.280 | 0.191 | 0.111 | -0.000 | -0.000 | 0.000 |

**Table 29:** Full factor regression ledger, including SPY, all eight underlying equities, VX front returns, VIX changes, and VVIX changes.

| Exclusion | Remaining underlyings | Option assets | Ann. return | Ann. vol | Sharpe | Sharpe CI lo | Sharpe CI hi | Sortino | Calmar | Omega | Info. ratio | Net delta | Beta SPY |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| All underlyings | 8 | 54 | 1.246 | 0.907 | 1.374 | 0.601 | 2.209 | 3.385 | 2.034 | 2.784 | 1.200 | 3.307 | 1.297 |
| No AAPL | 7 | 48 | 1.253 | 0.949 | 1.321 | 0.571 | 2.091 | 3.233 | 2.016 | 2.673 | 1.147 | 3.216 | 1.240 |
| No AMZN | 7 | 48 | 1.286 | 0.920 | 1.397 | 0.578 | 2.242 | 3.710 | 2.097 | 2.899 | 1.223 | 3.344 | 1.178 |
| No GOOGL | 7 | 49 | 1.214 | 0.895 | 1.357 | 0.547 | 2.213 | 3.325 | 1.936 | 2.740 | 1.180 | 3.246 | 1.265 |
| No JPM | 7 | 49 | 1.202 | 0.917 | 1.311 | 0.650 | 2.015 | 3.032 | 2.540 | 2.741 | 1.106 | 3.186 | 1.105 |
| No META | 7 | 47 | 1.250 | 0.944 | 1.323 | 0.566 | 2.149 | 3.188 | 2.011 | 2.658 | 1.163 | 3.381 | 1.375 |
| No MSFT | 7 | 49 | 1.410 | 0.955 | 1.476 | 0.750 | 2.228 | 3.807 | 2.386 | 3.053 | 1.312 | 3.406 | 1.627 |
| No NVDA | 7 | 46 | 1.085 | 0.821 | 1.322 | 0.547 | 2.151 | 3.256 | 1.893 | 2.750 | 1.157 | 3.571 | 0.775 |
| No TSLA | 7 | 47 | 1.198 | 0.842 | 1.424 | 0.671 | 2.251 | 3.612 | 1.988 | 3.118 | 1.265 | 3.330 | 1.287 |
| No META/NVDA/TSLA | 5 | 32 | 0.934 | 0.837 | 1.115 | 0.336 | 1.940 | 2.279 | 1.407 | 2.352 | 0.968 | 4.019 | 1.217 |

**Table 30:** Leave-one-underlying-out robustness for the VIX-enabled Greek Markowitz option portfolio.

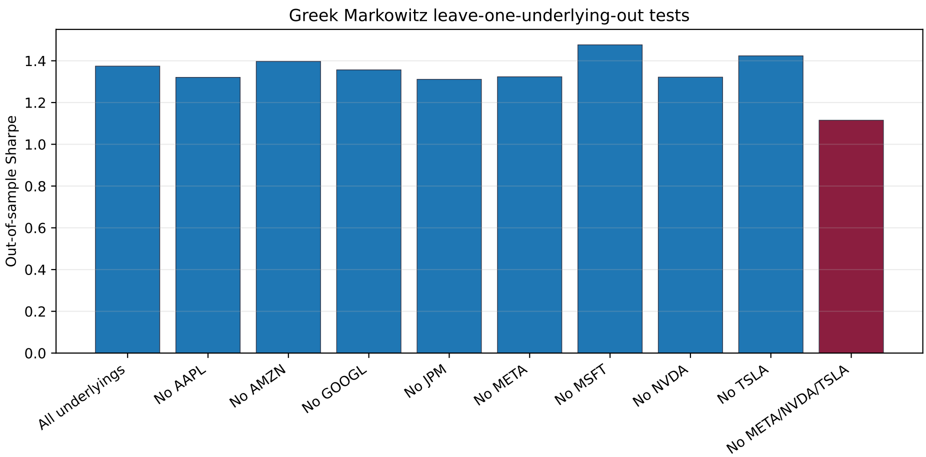


**Figure 5:** Out-of-sample Sharpe ratios for leave-one-underlying-out variants.

| Diagnostic | Value |
|---|---|
| Rolling 36M OOS months | 20.000 |
| Rolling 36M OOS ann. return | 1.460 |
| Rolling 36M OOS ann. vol | 0.882 |
| Rolling 36M OOS Sharpe | 1.656 |
| Rolling 36M OOS Sortino | 2.983 |
| Rolling 36M OOS Calmar | 2.561 |
| Rolling 36M OOS Omega | 3.307 |

**Table 31:** Supplemental rolling 36-month OOS diagnostic.

### *A.4. Instrument and settlement convention reference*

Every candidate option is first translated into a cashflow convention. Equity options record premium, underlying, strike, expiry, right, exercise style, mark, Greek model, payoff source, and split adjustment. VIX options record OSI symbol, VX-forward anchor, VIX settlement source, Black–76 Greek model, and whether expiry P&L used exact VRO/SOQ or a settlement proxy. Short options additionally record assignment-risk flags, simulated margin funding, and short-premium budget use.

A weak implementation-quality score does not automatically exclude a contract. It widens the haircut, raises the stress or capacity penalty, or downgrades the claim. This is why the paper can show VIX option diagnostics without treating proxy settlement as exact listed settlement.

### *A.5. Proof sketches, solver reductions, and implementation notes*

The covariance identity follows from the factor representation. If $r_O = \mu_O + Bf + \varepsilon$, $\mathrm{Cov}(f, \varepsilon) = 0$, $\Omega = \mathrm{Var}(f)$, and $\Sigma_\varepsilon = \mathrm{Var}(\varepsilon)$, then

$$\mathrm{Var}(r_O) = B\Omega B^\top + \Sigma_\varepsilon.$$

The matrix is positive semidefinite because both terms are positive semidefinite. This is the option-risk-model analogue of an equity factor covariance model.

The tangency result is the standard Markowitz argument. Sharpe is homogeneous of degree zero, so impose $q^\top \Sigma_O q = 1$ and maximize $q^\top \mu_O$. The Lagrangian first-order condition gives $\mu_O = 2\lambda\Sigma_O q$, hence $q \propto \Sigma_O^{-1}\mu_O$. Constraints replace this closed-form direction with a fractional program over the feasible set.

The reproducibility package exposes the model code, empirical pipeline, publication tables, figures, strategy-return ledgers, factor regressions, attribution, regime splits, VIX settlement records, claim-audit records, and `tables/empirical_summary.json`. The audit layer independently checks data lineage, point-in-time timing, optimizer constraints, output reproducibility, figure visibility, bibliography scope, caveat text, and LaTeX rendering.

| Moment | Value |
|---|---|
| Median residual mean | 0.076 |
| Median residual vol | 2.076 |
| Median absolute residual autocorr(1) | 0.098 |
| Systematic covariance rank | 27.000 |

**Table 32:** Training-window residual diagnostics for the local Greek-factor approximation.

The production gap remains visible. Exact VRO/SOQ settlement, executable option quotes, calibrated costs, margin, assignment, borrow, capacity, order routing, and broker reconciliation would be required for production claims. Those claims are not made in this article. The current contribution is the option-only theory object, a point-in-time research implementation, and an audit layer that prevents stronger wording when the data do not support it.